

claude-windows / 068560d3-1ca2-4387-981a-d2d6bca55414

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf_b3a30c42-346\agent-aab91c5031f4f8502.jsonl`  
- SHA-256: `e64b612c214b808f7650fda6e08672af17f8bf1f922b454343a0f4b1d80b2623`  
- Source modified: `2026-06-16T02:28:01+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_b3a30c42-346`  
- session_id: `068560d3-1ca2-4387-981a-d2d6bca55414`
```

Transcript

1. user (2026-06-16T02:24:52.618Z)

Read these docs IN FULL and summarize the DESIGNED (and any built) golden-set storage + sharing + regression model: docs/GOLDEN_DATA_SHARING.md, docs/GOLDEN_SET_IMPLEMENTATION.md, docs/GOLDEN_SET_PHASE1_VIDEOS.md, docs/GOLDEN_SET_VENDORING.md, docs/EVALUATION.md, docs/EXPERIMENT_HARNESS.md, docs/ABLATION_LAYER.md, docs/ROORA_LABELING_GUIDELINES.md, docs/QUALITY_ITERATIONS.md. ANSWER: (1) How are golden labels stored/shared today (format, location, video_id keying)? (2) CRUX: does any doc ALREADY propose committing structured per-video artifacts to the repo for video-free regression? If so summarize it precisely ? we must not duplicate or contradict it, only extend it. (3) The labeling schema (Roora) and its versioning. (4) What's built vs planned vs owner-gated. (5) Stated constraints on sharing golden data (licensing, privacy, minors, repo). Cite doc names/sections.

2. assistant (2026-06-16T02:24:59.507Z)

I'll read all the requested docs in full. Let me start by checking which exist and reading them.

3. user (2026-06-16T02:25:01.834Z)

Already up to date.

---FILES---

```
-rw-r--r-- 1 avidu 197609 3926 Jun 15 14:11 docs/ABLATION_LAYER.md  
-rw-r--r-- 1 avidu 197609 7036 Jun 15 14:11 docs/EVALUATION.md  
-rw-r--r-- 1 avidu 197609 15952 Jun 15 14:11 docs/EXPERIMENT_HARNESS.md  
-rw-r--r-- 1 avidu 197609 4025 Jun 15 14:11 docs/GOLDEN_DATA_SHARING.md  
-rw-r--r-- 1 avidu 197609 6240 Jun 2 22:52 docs/GOLDEN_SET_IMPLEMENTATION.md  
-rw-r--r-- 1 avidu 197609 8513 Jun 4 10:10 docs/GOLDEN_SET_PHASE1_VIDEOS.md  
-rw-r--r-- 1 avidu 197609 16664 Jun 13 22:54 docs/GOLDEN_SET_VENDORING.md  
-rw-r--r-- 1 avidu 197609 27728 Jun 14 16:35 docs/QUALITY_ITERATIONS.md  
-rw-r--r-- 1 avidu 197609 30436 Jun 15 14:11 docs/ROORA_LABELING_GUIDELINES.md
```

4. assistant (2026-06-16T02:25:04.751Z)

All nine docs exist. Let me read them in full, starting with the most directly relevant to the question (storage/sharing/regression).

5. user (2026-06-16T02:25:05.712Z)

```
1      # Golden / GPU-Extractor Data Sharing ? GPU box ? dev sessions  
2  
3      **Status:** convention + plan (2026-06-15). Tracked in **#175**.  
4      **Purpose:** a single cloud-synced location for the GPU-extracted golden artifacts  
(trajectory CSVs, `_golden_features.json`, manifests, baseline snapshots) so the **GPU box**  
and **CPU dev/agent sessions** share the same files ? without committing large data to git.  
5
```

```

6      ## Why this exists
7      The rally-quality measurement is split across two machines by capability:
8      - **`backend/eval/fusion_golden.py`** (shuttle+person feature **extraction**) needs the
**videos**, the **trajectory CSVs**, the **`.rallies.csv` golden labels**, and a **GPU/WSL**
stack ? only the GPU box can run it.
9      - **`backend/eval/fusion_compare.py`** / **`backend/eval/ablation.py`** (the LOVO
**litmus**: ?F1, bootstrap CI, paired sign test) are **pure-CPU re-scoring** that read only the
small `*_golden_features.json` ? any session (incl. a CPU dev container) can run them.
10
11     Today those artifacts sit in gitignored `output/` (per-machine) and scattered local
folders (`...\Annotation Setup\Collect\Trajectories\`, `...\Golden Labelled\`). Nothing syncs
them between machines, so a CPU session cannot see the GPU box's extraction. This wires that gap
with OneDrive.
12
13     ## Canonical shared folder
14     `%USERPROFILE%\OneDrive\rally-annotator\golden-data\` ?
`C:\Users\avidu\OneDrive\rally-annotator\golden-data\`
15
16     OneDrive is present + signed-in on both the GPU box and the dev machine under the same
account, so it auto-syncs. Layout (date/size-tagged so a re-extraction is a new folder, never an
overwrite):
17
18     ```
19     golden-data/
20         README.md
21         features/<corpus-tag>/          *_golden_features.json      (small ? the key shared
artifact)
22         trajectories/<corpus-tag>/      *_predict.csv / *_wasb.csv (medium ? reusable across
runs)
23         manifests/                     *_lovo_manifest.json
24         baselines/                     leaderboard / regression snapshots (or keep tracked in
eval_baselines/)
25     ```
26
27     `<corpus-tag>` = e.g. `2026-06-15-n6` (date + corpus size). **Do NOT** put raw videos
here (tens of GB) ? keep those local to the GPU box.
28
29     ## The GPU ? CPU workflow
30     1. **GPU box** (videos + CSVs + labels present):
31         ```
32         python -m backend.eval.fusion_golden --manifest <manifest> \
33             --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-
data\features\2026-06-15-n6"
34         ```
35         (or extract to local `output/` then copy the `*_golden_features.json` into the shared
folder).
36     2. OneDrive syncs the JSONs to the dev machine automatically (small files, seconds).
37     3. **CPU dev/agent session** points the analysis at the shared folder:
38         ```
39         python -m backend.eval.fusion_compare --features-dir "...golden-
data\features\2026-06-15-n6" --record
40         python -m backend.eval.ablation      --features-dir "...golden-
data\features\2026-06-15-n6" --granularity group
41         ```
42         ? real person-feature ?F1 / CI / sign test. Copy the JSONs into the repo's `output/`
to also light up the `skipif` golden-litmus tests
(`tests/test_ablation.py::test_litmus_reproduces_gate_a`, `tests/test_fusion_compare.py`).
43
44     ## Optional code ergonomics (in #175, not required)
45     - A `RALLY_GOLDEN_DIR` env var (default `output/`) honored by `fusion_golden` /
`fusion_compare` / `ablation` so neither machine hardcodes the path. (The CLI `--out-dir` /
`--features-dir` flags already make this a convention; the env var is just ergonomics.)
46     - A tiny `scripts/sync_golden.py` to copy `output/*_golden_features.json` ? the shared
folder.
47

```

```

48     ## Guardrails
49     - The shared folder lives **outside the repo** ? never committed. Keep `output/`
gitignored and do **not** symlink the shared folder into a tracked path.
50     - The features derive from the **Proprietary** corpus ? keep the OneDrive folder
**private** (no public share links).
51
52     See also: `docs/NEXT_STEPS.md` (the "Next-session pickup" anchor),
`docs/MULTI_SIGNAL_FUSION_PLAN.md` (P3), `docs/TRACKNET_WSL_SETUP.md` (the GPU/WSL detector
setup).
53

```

6. user (2026-06-16T02:25:06.205Z)

```

1     # Golden-Set Subproject ? Design & Implementation Overview
2
3     **Status:** In progress . **Date:** 2026-05-30
4     **Companion docs:** design rationale + vendor shortlist in
[GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) (ADR-006); scoring mechanics in
[EVALUATION.md](EVALUATION.md).
5
6     This is the **implementation tracker** for turning the golden-set/vendor *design* into
shipped code. It's split into small, independently-reviewable PRs. The whole subproject is
**vendor-agnostic by construction** ? only one slice (GS-3, `HumanVendorProvider`) depends on
which vendor you pick offline; everything else can land first.
7
8     > **Update (2026-06-02):** Vendor selected for the **initial pilot ? Roora**
(Bengaluru). The **annotation guideline, intake-manifest template, and NDA** are authored in the
collector repo `docs/annotation/`, and the collector's `import-local` ingestion now covers the
**3 pilot sports** (badminton, pickleball, table tennis). On the indexer side, GS-1
(`file_provider.py`), GS-2 (`backend/eval/regression.py` + `run_regression.py`), and the GS-4
oracle scaffold (`automated_provider.py`) have landed; the remaining slice is the vendor-
dependent **GS-3 `HumanVendorProvider`**, parked until the Roora pilot confirms quality.
9
10    ---
11
12    ## Architecture in one picture
13
14    ```
15        video_ref ??? AnnotationProvider.submit()/poll()/fetch() ???
List[(start,end)]  ground truth
16
17        ?           ?           ?           ?
18        FileProvider HumanVendor Automated(oracle)
(GS-1) (GS-3) (GS-4)
19        backend.eval.metrics.score() ??? existing, unchanged
20
21        production pipeline ??? DB ??? backend.eval.harness.get_predictions() ???
List[(start,end)]  predictions
22
23        regress(video, tier) (GS-2)
24
25        compares the two
26
27        ```
28
29    ## PR breakdown
30
31    | Slice | Branch | Scope | Depends on | Vendor-dependent? |
32    |---|---|---|---|---|
33    | **GS-1** | `feat/annotation-provider-registry` | `AnnotationProvider` ABC + registry +
`FileProvider` + tests | ? | No |

```

```

34 | **GS-2** | `feat/regression-runner` | `regress(video, tier, tolerance)` over existing
harness + `baseline.json` snapshot/compare + CLI | GS-1 | No |
35 | **GS-3** | `feat/human-vendor-provider` | `HumanVendorProvider` (submit/poll/fetch
against the chosen vendor's API or file-drop), DPA/secure-link handling | GS-1, **your vendor
pick** | **Yes** |
36 | **GS-4** | `feat/automated-oracle-provider` | `AutomatedProvider` fast-tier oracle ?
**interface stub** (injected labeler + fake; concrete model = backlog) | GS-1, GS-2 | No |
37 | **GS-5** | `feat/regression-triggers` | Wire triggers: CI gate, on-demand CLI,
scheduled run | GS-2, GS-4 | No |
38
39 Each row is one PR. GS-1 ? GS-2 ? GS-4 ? GS-5 can proceed **now**, independent of vendor
selection. GS-3 is parked until you've chosen a vendor (then it's a thin adapter behind the GS-1
interface).
40
41 ---
42
43 ## Slice details
44
45 ### GS-1 ? Provider registry + FileProvider ? (this PR)
46 - `backend/annotations/base.py` ? `AnnotationProvider` ABC (`submit`/`poll`/`fetch` +
`annotate()` convenience) and `annotation_registry`, mirroring `backend/providers` and
`backend/indexer` registry patterns.
47 - `backend/annotations/file_provider.py` ? `FileProvider`: the "job" is a CSV that
already exists (formalizes today's manual/ShuttleSet flow); reuses
`eval.annotations.load_annotations`.
48 - Tests assert the registry contract **and** the keystone (provider output is directly
scorable by `metrics.score()`).
49 - **Design choice:** `fetch()`/`annotate()` raise loudly on a missing/incomplete job
rather than returning `[]`, so a broken label source can't silently masquerade as "0 rallies"
and skew a regression to green.
50
51 ### GS-2 ? Regression runner (next)
52 `regress(video, tier="fast", tolerance=0.05)`: get GT from
`annotation_registry.get(tier)`, read predictions via the existing `eval.harness`, score with
the existing `metrics`, compare against a per-video `baseline.json`, and flag a regression when
precision/recall drop beyond tolerance. Pure orchestration over existing parts + a small
baseline store. CLI: `python run_regression.py --video ... --tier file`.
53
54 ### GS-3 ? HumanVendorProvider (after your vendor pick)
55 A concrete provider that submits a (licensed/owned-only) video to the chosen vendor and
fetches the returned rally CSV. Carries the privacy controls from the design doc (secure time-
limited links, delete-on-delivery, DPA). Thin adapter ? the interface already exists from GS-1.
56
57 ### GS-4 ? AutomatedProvider (fast-tier oracle) ?? oracle rule
58 An independent strong labeler for the fast/CI tier. **Must be a different model lineage
than whatever production uses** (else shared blind spots ? false green). A fast-tier failure is
a *smoke alarm* that escalates to the GS-3 human tier, never an authoritative verdict on its
own.
59
60 ### GS-5 ? Regression triggers
61 Wire `regress()` into CI (pre-merge gate on pipeline changes), an on-demand CLI (the
"surprise regression"), and an optional scheduled run. Includes a deliberate re-snapshot path so
legitimate production improvements can rebaseline instead of raising false regressions.
62
63 ---
64
65 ## Conventions
66 - **Branch per slice**, one PR each, tests green before PR, full suite must stay green.
67 - New runtime deps: **none** through GS-2 (GS-3 may add a vendor SDK; GS-4 may reuse the
existing `google-genai` for a Gemini oracle).
68 - All ground-truth video stays **licensed/owned** ? see the privacy section of the
design doc.
69

```

```

1      # Phase-1 Golden-Set Video Collection Spec
2
3      **Status:** Design v1 . **Date:** 2026-06-02
4      **Companion docs:** [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md) (vendor process,
ADR-006) . [VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md) (capture spec the
user-facing guide derives from) . scoring in [EVALUATION.md](EVALUATION.md).
5
6      What footage to gather for the **first round** of golden-set creation so the trained
rally segmenter reaches good precision/recall ? and so the eval number actually predicts
production. Golden-set creation is **iterative**: expect multiple rounds, expanding toward
whatever axis round-1 metrics show is weakest.
7
8      ---
9
10     ## 1. The principle that dominates everything: match the deployment distribution
11
12     The CV literature is consistent: **models trained on broadcast footage fail to
generalize to self-recorded amateur video** ? broadcast uses a "straight, fixed camera with
minimum variability," while real non-elite footage has limited coverage, occlusions, and
inconsistent viewpoints ([team-sports
survey](https://link.springer.com/article/10.1007/s10462-024-10934-9), [unstructured-video event
detection](https://arxiv.org/pdf/2002.08097)). This is the **same failure mode as ADR-004**
(broadcast cameras *track players*, so player pixel-motion inverts during rallies).
13
14     Our product ingests **users' own match recordings** (a fixed phone/GoPro on a tripod).
Therefore:
15
16     - The golden set ? **especially the held-out eval split** ? must be dominated by
**static, self-recorded, full-court** footage that mirrors real users.
17     - **ShuttleSet / broadcast** footage is fine as *training* augmentation and as a
contrast set, but must be **minimal in eval**. Measuring precision/recall on broadcast yields a
number that does **not** predict production.
18
19     ---
20
21     ## 2. Primary recording profile (also the basis of the user-facing capture guide)
22
23     Grounded in the badminton-CV setup literature (camera ~2 m behind the baseline, ~4.5 m
elevated, ~30° down-tilt, whole court + headroom above the net ?
[source](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)):
24
25     | Setting | Spec | Why |
26     |---|---|---|
27     | Mount | Fixed tripod; **no pan/zoom** during play | Avoids the ADR-004 camera-tracking
trap |
28     | Position | Behind one baseline/end, centered | Full-court, symmetric view |
29     | Height | **3?5 m** elevated (~4.5 m ideal) | De-occludes the far player & net |
30     | Tilt | **~30° down** | Optimal coverage, low keystoneing |
31     | Framing | Entire court **+ headroom above the net** ; players never leave frame |
Detection + court-relevance need it |
32     | Lighting | Even; no backlight/glare | Reduces blur & false motion |
33
34     Most of round 1 should use this profile; a minority deliberately varies it (§4) to probe
robustness.
35
36     ---
37
38     ## 3. Resolution & frame rate (per sport)
39
40     A controlled study: table-tennis ball detection was **22.5% @ 30 fps ? 93.2% @ 60 fps**
(120 fps ? 60 fps) ([Nature Sci Rep](https://www.nature.com/articles/s41598-023-42966-6));
OpenTTGames records at 120 fps. The ball/shuttle is the camera-invariant rally signal (our WASB
substrate, ADR-001), so fps is **not** a nicety for fast sports.
41
42     | Sport | Min resolution | Recommended fps | Note |

```

```

43      |---|---|---|---|
44      | Badminton | 1080p (720p floor) | 30 ok, **50/60 better** | Shuttle decelerates fast;
60 sharpens smashes |
45      | Pickleball | 1080p | **30?60** | Ball slower than TT; 30 likely fine, 60 safer |
46      | Table tennis | 1080p | **60 min (120 ideal)** | 30 fps collapses ball detection |
47
48      > These are **literature-backed starting points**; the published minimums get finalized
empirically against our golden set (VIDEO_RECORDING_GUIDELINES.md §3).
49
50      ---
51
52      ## 4. The Phase-1 diversity matrix
53
54      Cover the **high-impact axes** now; don't chase rare conditions in round 1. "Guards" =
whether the axis mostly protects **P**recision (not calling dead-time a rally) or **R**ecall
(not missing rallies).
55
56      | Dimension | Vary across? | Guards | Round-1 target |
57      |---|---|---|---|
58      | **Sport** | badminton, pickleball, table tennis | both | all 3 |
59      | **Format** | singles **and** doubles (badminton, pickleball) | R | both for those 2 |
60      | **Camera profile** | mostly primary-static; 1?2 side-elevated; 1?2 lower/phone-height;
**1?2 broadcast (contrast only)** | both | ~80% static / ~20% variants |
61      | **Venue / court color** | wood, green, blue synthetic | P | ?3 venues |
62      | **Lighting** | bright gym, dim club, mixed/backlit | both | ?3 conditions |
63      | **Background clutter** | empty hall vs people moving behind court | **P** (crowd
motion ? false positives) | both |
64      | **Players** | skill, attire colors, body types, L/R-handed, skin tone/age | R |
spread; don't reuse one pair |
65      | **Rally edge cases** | service faults, lets/replays, 1?2-shot & long rallies, net-
cords, coaching pauses, warm-up | boundary **P/R** | ensure present |
66      | **Resolution/fps ablation** | same match re-encoded 480/720/1080p; 30 vs 60 fps |
calibration | 1 matched set/sport |
67
68      ---
69
70      ## 5. Split it so the precision/recall number is honest (no train/eval leakage)
71
72      The #1 video-ML mistake: the **same match in train and eval inflates the score** ([data-
leakage guide](https://towardsdatascience.com/data-science-mistakes-to-avoid-data-
leakage-e447f88aae1c/)).
73
74      - **Split by match / recording session ? never by clip.** All clips from one match go
entirely to train *or* eval.
75      - **Hold out whole *conditions*** (a venue, a camera profile, a player pair the model
never saw) so eval measures generalization, not memorization.
76      - Reserve **~? of each sport as a deployment-like held-out eval set** (static, self-
recorded). Broadcast/ShuttleSet may enter train but should be **minimal in eval**.
77
78      > **Guardrails ? TODO (design with the user):** train and eval live in **separate
catalogs** today, kept apart by convention. We want hard guardrails ? permissions / explicit
approval / re-confirmation at ingest & export ? so a video can **never** silently land in both.
Tracked in BACKLOG.md; to be designed in a later session.
79
80      ---
81
82      ## 6. Concrete "round 1" shopping list (per sport)
83
84      Aligns with the 12?15-videos/sport scale plan; the 3-clip Roora calibration is a subset
of this.
85
86      - **8 static-primary** clips spanning venues / lighting / skill (?4 singles, 4 doubles
where applicable)
87      - **2 camera-variant** clips (side-elevated, lower height) ? robustness probe
88      - **1 resolution/fps ablation set** (one match @ 480/720/1080p, 30 & 60 fps)

```

```

89     - **1 broadcast contrast** clip (ShuttleSet) ? *train/contrast only*
90     - Each ~274 min, **deliberately including the edge cases** above ? **~12 clips/sport**;
hold out **4 whole matches** for eval.
91
92     ---
93
94     ## 7. Deliberately defer to later rounds
95
96     Outdoor courts, phone-handheld/jitter, extreme angles, 4K, unusual net/court variants,
crowd-heavy tournaments. Add these once round-1 precision/recall shows the weakest axis, then
expand toward it (find the quality "knee" ? VIDEO_RECORDING_GUIDELINES.md §3).
97
98     ---
99
100    ## 8. Two cautions specific to precision (the stated goal)
101
102    - **Backgrounds with people moving** behind the court are the biggest false-positive
driver. Include some so the model learns to reject them ? and make sure the annotation
guideline's "ignore all dead time" rule is airtight.
103    - **Warm-up / knock-up / practice swings** look like rallies. Capture some, and confirm
the vendor **excludes** them (the Roora guide now has an explicit rule + a how-to-spot-it tip).
Early precision is bounded more by **guideline / inter-annotator consistency (IAA)** than by raw
data volume.
104
105    ---
106
107    ## 9. Sources
108
109    - [Team-sports action-recognition survey ? Springer
2024](https://link.springer.com/article/10.1007/s10462-024-10934-9)
110    - [Unstructured-video event detection ? arXiv
2002.08097](https://arxiv.org/pdf/2002.08097)
111    - [Badminton service-height CV setup (camera 4.5 m / 30°) ?
PMC](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)
112    - [VideoBadminton dataset ? arXiv 2403.12385](https://arxiv.org/html/2403.12385v1)
113    - [Table-tennis frame-rate vs detection accuracy ? Nature Sci
Rep](https://www.nature.com/articles/s41598-023-42966-6)
114    - [OpenTTGames (120 fps) dataset](https://lab.osai.ai/)
115    - [Action-spotting / segmentation survey ? arXiv
2505.03991](https://arxiv.org/html/2505.03991v2)
116    - [Data-leakage in train/test splits](https://towardsdatascience.com/data-science-
mistakes-to-avoid-data-leakage-e447f88aae1c/)
117

```

8. user (2026-06-16T02:25:09.628Z)

```

1     # Golden-Set Generation & Vendor Process (v1)
2
3     **Status:** Design v1 . **Date:** 2026-05-30 . **Vendor shortlist added 2026-05-30
($6)**
4     **Owner decisions captured:** licensed/owned video only . rally `[start,end)` depth .
tiered turnaround (fast automated + slow human) . vendor bake-off . **Bangalore/India-local
vendor preference**
5
6     This document defines how we produce **golden-set ground truth** (hand-verified rally
boundaries) for the badminton highlights product, how we **select annotation vendors**, and how
that same machinery powers **on-demand / surprise regression testing** of the production tool.
7
8     > **Companion docs:** the scoring mechanics live in [EVALUATION.md](EVALUATION.md); the
annotated-corpus rationale is [ADR-002](archives/decisions/DECISIONS.md); the decision record
for *this* process is **ADR-006**. Video-capture quality requirements (a dependency of
trustworthy golden sets) are tracked separately in
[VIDEO_RECORDING_GUIDELINES.md](VIDEO_RECORDING_GUIDELINES.md). The `regress()` +
`baseline.json` machinery in §5 is generalized into a **variant A/B + promotion gate** in
[EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md).

```

9

10 > ****Update (2026-06-02):**** S6 shortlist done; the founder selected ****Roora**** (Bengaluru) for the ****initial paid pilot**** ? a ****single-vendor pilot first****, not the full bake-off below. Roora's site/capability/location are verified; ****quality is TBD via the pilot****. ****Step 0** (the annotation guideline) is authored **** ? collector repo `docs/annotation/` (`ROORA\_BRIEF.md` rally CSV schema + `ANNOTATION\_GUIDE.md` illustrated guide + intake-manifest & NDA templates). Pilot scope is **3 sports** (badminton, pickleball, table tennis); the collector's `import-local` ingestion for all three is built. Still pending: the internal ****gold key**** and the vendor-dependent ****GS-3 `HumanVendorProvider`**** (GS-1/GS-2/GS-4 already landed ? see GOLDEN_SET_IMPLEMENTATION.md).**

11

12 ---

13

14 **## 1. The keystone insight: one scorer, three jobs**

15

16 A vendor's golden-set output is a list of rally `[start,end)` intervals ? ****structurally identical to a prediction**** from our pipeline. So the ***exact same*** deterministic scorer in `backend/eval/metrics.py` (`interval\_iou` ? `match\_intervals` ? `score`) grades all three of:

17

18 1. ****Product vs. ground truth**** ? the precision/recall/F1/IoU we already report (`EVALUATION.md`).

19 2. ****Vendor vs. our internal answer key**** ? the quality gate during vendor selection.

20 3. ****Annotator vs. annotator**** ? inter-annotator agreement (IAA), our proxy for "is this even a well-defined task?"

21

22 This symmetry is what makes the whole system small, auditable, and extensible. ****No new scoring code is needed**** ? only a new ***source*** of intervals (the vendor).

23

24 ---

25

26 **## 2. Why this also solves the privacy nuance**

27

28 The monetization goal (hosted API) carries a "someone views the user's video" concern. This process sidesteps it ****by construction****:

29

30 - ****Golden sets are built only from footage we license or own**** ? starting with the founder's ****constantly-growing personal collection**** of match videos, plus ShuttleSet (ADR-003) and any purpose-recorded clips. ****No end-user uploads ever flow to a vendor.****

31 - Regression tests therefore run against a ****representative held-out corpus****, not literal user content.

32 - All video egress to any third party passes through a ****single provider boundary**** (§4), so the security controls (DPA, no-reuse, delete-on-delivery, access logging) live in exactly one place.

33

34 > ****Honest tradeoff:**** a licensed/owned corpus may not perfectly mirror real users' cameras, courts, and lighting. A green regression on our corpus is therefore ***strong evidence***, not a ***guarantee***, for production traffic. Revisit if field behavior diverges from corpus behavior ? and see VIDEO_RECORDING_GUIDELINES.md, which aims to shrink that gap by steering users toward capture conditions our corpus represents.

35

36 ---

37

38 **## 3. Selecting vendors ? the bake-off**

39

40 **### Step 0 ? Write the annotation guideline (highest-leverage artifact)**

41 Before contacting anyone, write a precise, example-rich definition of ****what a rally is****:

42 - ****Start trigger:**** e.g. first serve motion vs. shuttle leaving the racket ? pick one, show clips.

43 - ****End trigger:**** shuttle hits floor/net/out, or umpire call.

44 - ****Edge cases:**** lets/replays, service faults, between-point walking, coaching pauses, mid-rally net-cord, broadcast replays and graphics.

45

46 Rally boundaries are genuinely subjective; ****ambiguity here is the #1 cause of bad golden sets.**** This guideline belongs conceptually with the per-sport `SportProfile`, so it

generalizes to tennis/cricket later.

```
47
48     > **Done (2026-06-02):** authored as the Roora **brief + illustrated guide** in the
collector repo `docs/annotation/`. Defines the rally CSV schema and the three golden rules ?
ignore all dead time; include **every** rally (incl. service faults & lets); a shot = one
racket/paddle/bat contact ? plus an `ending_reason` taxonomy (winner / forced_error /
unforced_error / service_fault / let / other, shared across net-separated sports) and per-sport
detail incl. **singles & doubles** for badminton & pickleball.
49
50     ### Step 1 ? Build a small internal gold key
51     One trusted expert (initially: the founder) hand-annotates **5?10 licensed clips** to
near-perfect. This is the answer key vendors are graded against ? kept private during the bake-
off.
52
53     ### Step 2 ? Run the bake-off (parallel pilot)
54     Send the **same** pilot clips to **2?3 shortlisted vendors** (see §6, favoring
Bangalore/India), withholding the key. Grade every submission with **our own harness**:
55
56     | Dimension | How we measure it | Tool |
57     |---|---|---|
58     | **Accuracy** | vendor CSV vs. gold key ? P/R/F1, mIoU, boundary error | `eval.score()`
|
59     | **Consistency** | 2 annotators per vendor ? pairwise IoU (IAA) | `interval_iou` |
60     | **Cost** | quoted ? or $ per video-minute | vendor quote |
61     | **Turnaround** | actual vs. promised, per tier | observed |
62     | **Ops friction** | file-drop vs. API; compliance with our CSV format |
`annotations.load_annotations()` |
63     | **Security** | DPA signed; DPDP/ISO posture; delete-on-delivery | contract review |
64
65     ### Step 3 ? Decide on measured quality-per-dollar
66     The bake-off numbers **are** the decision ? not sales decks. Low IAA at a vendor means
either a weak vendor **or** a vague guideline (Step 0); investigate before blaming the vendor.
67
68     ### Step 4 ? Onboard the winner(s)
69     Sign DPA + the security controls in §4; codify the guideline as their spec; agree SLAs
for both tiers (§5).
70
71     ---
72
73     ## 4. The extensible architecture ? an `AnnotationProvider` registry
74
75     This codebase already uses pluggable registries (validators, segmenters, AI providers,
sports). The idiomatic move is the same shape for annotation sourcing:
76
77     ```python
78     class AnnotationProvider(ABC):
79         def submit(self, video_ref, guideline, sport) -> str:           # returns job_id
80             ...
81         def poll(self, job_id) -> str:                                   #
queued|running|done|failed
82             ...
83         def fetch(self, job_id) -> List[Interval]:                     # canonical
[(start,end), ...]
84             ...
85     ```
86
87     `fetch()` returns the **same `List[Interval]** that `annotations.load_annotations()`
already produces, so everything downstream is unchanged.
88
89     Three implementations cover today plus both tiers:
90
91     | Provider | Role | Tier |
92     |---|---|---|
93     | `FileProvider` | Reads a pre-made CSV (formalizes today's manual / ShuttleSet flow) |
? |
```

```

94      | `HumanVendorProvider` | The bake-off winner; authoritative ground truth, corpus
growth, deep audits | **Slow (days)** |
95      | `AutomatedProvider` | An independent strong model labels rallies programmatically |
**Fast (hours)** |
96
97      The tiers are just two providers behind one interface, selectable per call.
98
99      ---
100
101     ## 5. The regression automation (the explicit ask)
102
103     > "given a video, get a golden set created and verify its precision and recall against
the production tool"
104
105     This becomes a thin wrapper over machinery that **already exists**:
106
107     ```python
108     def regress(video, tier="fast", tolerance=0.05):
109         gt          = annotation_registry.get(tier).fetch(          # NEW: the provider
110                        annotation_registry.get(tier).submit(video, GUIDELINE, sport))
111         predictions = process(video)                                # existing production
pipeline ? DB
112         s          = eval.score(predictions, gt)                    # existing metrics.py,
unchanged
113         regressed  = (s.recall < baseline[video].recall - tolerance or
114                        s.precision < baseline[video].precision - tolerance)
115         if regressed: alert(video, s, baseline[video])
116         return s
117     ```
118
119     Only **one new component** (the provider). `run_eval.py`, `harness.py`, and `metrics.py`
are reused verbatim. Snapshot a `baseline.json` per corpus video so "regression" is a measured
delta, not a vibe.
120
121     **Triggers** (all feed the same `regress()`): CI on pipeline changes · on-demand (the
"surprise" case) · scheduled cron · pre-release gate.
122
123     ### ?? The oracle problem (a real correctness trap)
124     The fast-tier `AutomatedProvider` **must be a different model lineage than production**.
If production becomes the offline ActionFormer segmenter (ADR-004) and the oracle is **also**
ActionFormer, they share blind spots ? the test shows green while both are wrong. Pairings that
stay honest:
125
126     - production = offline learned segmenter ? fast-tier oracle = **paid Gemini full-video**
(different lineage).
127     - production = Gemini hybrid ? fast-tier oracle = a **different VLM** or the slow human
tier.
128
129     Treat the fast tier as a **smoke alarm**, not ground truth: any fast-tier "regression"
**escalates to the slow human tier** for confirmation before it blocks a release.
130
131     ---
132
133     ## 6. Vendor shortlist (Bangalore/India-favoring)
134
135     > **Researched 2026-05-30 via a lean manual pass** (targeted web search + direct
site/source verification ? NOT the heavy automated workflow, which failed). **Confidence is
labelled per row.** **No pricing is listed because none could be verified** ? every vendor in
this space quotes per-project, so ?/clip is an offline-conversation item (see §6.3). Treat
single-source rows as **leads to verify on the call**, not vetted recommendations.
136
137     ### 6.1 Tier-1 candidates ? verified India presence + video capability (talk to these
first)
138
139     | Vendor | Location (verified) | Video / temporal labeling | Notes for the call |

```

Confidence |

140 |---|---|---|---|---|

141 | **iMerit** | **Bengaluru** office (A101 & A118, 43 Residency Rd, 560025; tech hub opened Oct 2022); also Kolkata HQ + US | Yes ? image/video/text/audio; explicit Sports & Biomechanics domain on their site | Largest & most enterprise-ready (5,500+ annotators); has a real **Bangalore** footprint for in-person coordination; ask for a sports/temporal-segmentation reference + a paid pilot. Higher-touch ? likely higher cost. | **High** |

142 | **TELUS International AI** (ex-**Playment**) | Playment was **Bangalore-founded** (2015); acquired by TELUS 2021, team retained | Yes ? built on Playment's 2D/3D image+video+LiDAR CV pipeline | Strong CV/video pedigree, Bangalore roots; but now a large global BPO ? confirm they'll take a **small startup pilot** and who the India delivery contact is. | **High** (origin) / **Med** (current India ops shape) |

143 | **Taskmonk** | **Bengaluru** | Yes ? "Video" listed as a labeling modality; platform **managed-service workforce** | Mid-size, India-local, does both tooling and service ? potentially the best **cost/agility** fit for a startup pilot. Verify temporal/event labeling specifically (their heritage is e-commerce catalog tagging). | **Med-High** |

144

145 **### 6.2 Tier-2 ? India video-annotation services, location/specialts to confirm on contact**

146

147 | Vendor | Stated location | Note | Confidence |

148 |---|---|---|---|

149 | **Inforks** | India (Kerala-based per third parties) | Targets startups/SMBs; image/video/text/audio; positions on cost-efficiency ? good cheap-pilot candidate | **Med** |

150 | **Analytics** | India + USA (since 2018) | General data-annotation incl. video; widely listed | **Med** |

151 | **Objectways** | HQ Scottsdale AZ; **India delivery in Coimbatore** (not Bangalore) | Does video/LiDAR; India delivery but not Bangalore-local | **Med** |

152 | **Roora** (Roora ML Pvt Ltd) | **Bengaluru** ? HSR Layout (verified, `rooragrp.com`) | Image/video/text annotation; sports industry listed. **Selected for the initial paid pilot (2026-06-02)** ? existence/capability/location verified; **quality TBD via the pilot**. | **Verified** (capability) · pilot pending |

153 | **Tika Data** / **Zuru** | Claimed **Bangalore** (single-source listicle each) | Smaller/startup annotation shops ? unverified beyond one directory mention; verify they exist & do temporal video before investing time | **Low** |

154

155 **### 6.3 Explicitly NOT verified / out of scope**

156 - **Pricing for everyone** ? all quote-only; do not assume. Get per-video-minute (or per-rally) quotes during the offline conversations.

157 - **Karya (Bengaluru)** ? confirmed local and reputable, but **speech/text/translation only, no video** ? **excluded** for our use case.

158 - **Sports/temporal track record** ? beyond iMerit's stated "Sports & Biomechanics" domain, vendors' specific **rally-boundary / temporal-event** experience is unconfirmed marketing copy; probe with the pilot (§3).

159 - **DPDP / DPA / ISO posture** ? not verified per-vendor; make it a checklist item in the bake-off (§3 Step 2 security row).

160

161 **### 6.4 Recommendation**

162 For the **offline conversations**, start with **iMerit** (Bangalore-local, enterprise QA, sports domain) and **Taskmonk** (Bangalore-local, likely more startup-friendly cost/agility) as the two primary contacts, with **TELUS/Playment** as a third if you want a heavyweight CV pedigree. Run the §3 bake-off across whichever 2/3 agree to a paid pilot, and decide on measured quality-per-rupee. **Verify every "Confidence: Med/Low" claim on the call before committing.**

163

164 > **Update (2026-06-02):** the founder opted to start with **Roora** (Bangalore-local, startup-friendly) as a **single-vendor pilot** rather than the iMerit/Taskmonk multi-vendor bake-off above. iMerit / Taskmonk / TELUS remain the fallback shortlist if the Roora pilot doesn't clear the §3/§4 quality bar.

165

166 > Sources (manual pass, 2026-05-30): iMerit Bengaluru hub ? imerit.net press / PRNewswire (Oct 2022) + contact listings; Playment/TELUS ? BusinessWire & Inc42 (2021 acquisition, Bangalore-founded); Taskmonk & Karya locations ? company sites + AnalyticsIndiaMag / IndiaAI; others ? AnalyticsIndiaMag / Ziloservices / vendor sites. Pricing intentionally omitted (unverifiable).

167

```

168    ---
169
170    ## 7. Suggested phasing
171
172    | Phase | Deliverable | Depends on |
173    |---|---|---|
174    | 0 | Annotation guideline + internal gold key (5?10 clips) | nothing ? can start now |
175    | 1 | Vendor bake-off ? pick winner(s) | $6 research + guideline |
176    | 2 | `AnnotationProvider` registry + `FileProvider` + `HumanVendorProvider` | winner
onboarded |
177    | 3 | `AutomatedProvider` (independent oracle) + `regress()` + baseline snapshots |
Phase 2 |
178    | 4 | Wire surprise triggers (CI / on-demand / scheduled) | Phase 3 |
179
180    Phase 0 is unblocked today and is the highest-leverage step regardless of vendor choice.
181
182    > Status (2026-06-02): Phase 0 guideline ? authored (collector `docs/annotation/`);
internal gold key still pending. Phase 1 is running as a single-vendor Roora pilot across
3 sports (badminton, pickleball, table tennis) rather than a multi-vendor bake-off. Phase
2's `FileProvider` (GS-1) and Phase 3's `regress()` + automated oracle (GS-2/GS-4) have already
landed; the main remaining slice is the vendor-dependent `HumanVendorProvider` (GS-3), with
trigger-wiring (GS-5) after it.
183
184    ---
185
186    ## 8. Open questions
187    1. Guideline authorship ? who owns the canonical rally definition, and is one
expert's key enough, or do we want 2-expert consensus for the answer key itself?
188    2. Corpus representativeness ? how well does the owned/licensed collection mirror
eventual user capture? (Ties directly to VIDEO_RECORDING_GUIDELINES.md.)
189    3. Baseline drift ? when production legitimately improves, how/when do we re-
snapshot `baseline.json` so old baselines don't raise false regressions?
190    4. Fast-tier oracle cost ? per-video Gemini cost at regression frequency (see
`archives/MONETIZATION_AUDIT.md` §4 for the per-video estimate).
191

```

9. assistant (2026-06-16T02:25:12.189Z)

Now the evaluation and harness docs, plus ablation, which detail the scoring and artifact formats.

10. user (2026-06-16T02:25:12.675Z)

```

1    # Annotation & Evaluation
2
3    How to measure how well the pipeline detects rallies, using hand-annotated
4    ground-truth timestamps. The harness scores predictions already stored in the
5    SQLite DB ? it makes no API calls and is fully deterministic, so you can run it
6    as often as you like for free.
7
8    ---
9
10   ## 1. What you annotate (and what it does / doesn't do)
11
12   You annotate, per video, the `[start, end)` time range of each rally. That's
13   all the evaluation harness needs.
14
15   > Important: these rally timestamps measure and tune rally segmentation
16   > ? i.e. where the pipeline says rallies begin and end. They do not train
17   > the shuttle detector (TrackNet/WASB), which learns from per-frame shuttle
18   > `(x, y)` coordinate labels, a different and much heavier annotation. Use rally
19   > timestamps as your golden evaluation set; only invest in frame-level
20   > coordinate labels if evaluation shows detection (not segmentation) is the
21   > bottleneck. See `docs/TRACKNET_WSL_SETUP.md`.
22

```

```

23  ## 2. Annotation file format (CSV)
24
25  One CSV per video, one rally per row:
26
27  ```csv
28  start,end
29  00:01:12,00:01:39
30  102.5,131.0
31  ```
32
33  - The `start,end` header is optional (auto-detected).
34  - Timestamps may be **decimal seconds** (`102.5`) or **colon-separated**
35    `MM:SS` / `HH:MM:SS` (`00:01:12`). Forms can be mixed.
36  - Blank lines and `#` comments are ignored.
37  - Malformed rows (bad timestamp, `start >= end`) fail loudly rather than being
38    silently skipped ? so labeling mistakes surface immediately.
39
40  Generate an empty template to fill in:
41
42  ```bash
43  python run_eval.py --scaffold --annotations rallies.csv
44  ```
45
46  ## 3. Running the evaluation
47
48  First make sure the video has been processed so its detections are in the DB
49  (the pipeline keys each video by its file MD5). Then:
50
51  ```bash
52  # Identify the video by file (its MD5 is computed for you)...
53  python run_eval.py --video "C:/videos/match.mp4" --annotations rallies.csv
54
55  # ...or by the stored video_id (MD5) directly:
56  python run_eval.py --video-id <md5> --annotations rallies.csv --stage both --iou 0.5
57  ```
58
59  Useful flags:
60
61  | Flag | Effect |
62  |-----|-----|
63  | `--stage {candidates,final,both}` | Which prediction stage(s) to score (default
64  `both`). |
65  | `--detector <name>` | Only score candidates from one detector (e.g. `yolo_hybrid`,
66  `tracknet_hybrid`). |
67  | `--iou <float>` | Match threshold (default `0.5`). |
68  | `--pr-curve` | Also report an IoU sweep (0.1?0.9). |
69  | `--show-failures` | List missed (FN) and spurious (FP) rallies with timestamps. |
70  | `--json <path>` | Write a machine-readable report. |
71  | `--db <path>` | Override the DB path (default: `output/<output.db_name>` from
72  `config.json`). |
73
74  ## 4. Reading the output
75
76  ```
77  =====
78  Rally evaluation ? video 'demo'
79  Ground-truth rallies: 3 | IoU threshold: 0.5
80  =====
81  Stage      Pred  TP  FP  FN  Prec  Rec  F1  mIoU  dStart  dEnd
82  -----
83  candidates    3   3   0   0  1.00  1.00  1.00  1.00  0.0s   0.0s
84  final         2   2   0   1  1.00  0.67  0.80  0.95  0.2s   0.5s
85  =====
86  [final] missed rallies (false negatives): 1
87  - 00:01:20?00:01:32

```

```

85     ``
86
87     - **TP / FP / FN** ? true positives (matched), false positives (spurious
88       predictions), false negatives (missed rallies).
89     - **Prec / Rec / F1** ? precision, recall, and their harmonic mean.
90     - **mIoU** ? mean temporal IoU over matched pairs (how tightly matched rallies
91       overlap their ground truth).
92     - **dStart / dEnd** ? mean absolute boundary error in seconds (are rallies
93       detected but trimmed in the wrong place?).
94
95     ### The key diagnostic: candidates vs. final
96
97     Scoring both stages tells you where to spend effort:
98
99     - **Candidates miss rallies the GT has** ? the **detector** is the bottleneck.
100       Improve the shuttle model (e.g. fine-tune TrackNet/WASB with frame-level
101       labels) or relax detection thresholds.
102     - **Candidates catch them but `final` is wrong/trimmed** ? the **segmentation /
103       AI handoff** is the bottleneck. Tune thresholds (`velocity_threshold`,
104       `dead_time_merge_gap`, `min_action_window_duration`) or the AI prompt ? no new
105       labels needed.
106
107     In the example above, detection is perfect (candidates: recall 1.0) but the AI
108     handoff dropped one rally (final: recall 0.67) ? so the fix is in segmentation,
109     not detection.
110
111     ## 5. How matching works
112
113     Predictions and ground truth are sets of `[start, end)` intervals. A prediction
114     **matches** a ground-truth rally when their **temporal IoU** ( $\text{overlap} \div \text{union}$ ) is
115     at least the threshold. Matching is **greedy and one-to-one** ? highest-IoU pairs
116     are claimed first, and each prediction/GT is used at most once, so a single
117     detection can't be credited for two rallies. Unmatched predictions are false
118     positives; unmatched ground-truth rallies are false negatives.
119
120     ## 6. Code map
121
122     | File | Role |
123     |-----|-----|
124     | `backend/eval/metrics.py` | Pure interval math: IoU, matching, P/R/F1, boundary error.
125     |
126     | `backend/eval/annotations.py` | Load/validate the ground-truth CSV. |
127     | `backend/eval/harness.py` | Read predictions from the DB, score, format the report. |
128     | `run_eval.py` | CLI entry point. |
129     | `tests/test_eval_*.py` | Deterministic tests for all of the above. |
130
131     ## 7. Troubleshooting
132
133     | Symptom | Cause & fix |
134     |-----|-----|
135     | `ERROR: database not found at output/...` | The harness only reads stored predictions
136       ? it does **not** run the pipeline. **Process the video first** (e.g.
137       `scripts/run_process_and_stitch.py`) so detections land in the DB, then re-run. |
138     | Every rally shows as a miss (FN), `Pred` is 0 | This video hasn't been processed, or
139       you used the wrong `video_id`. The pipeline keys videos by **file MD5** ? pass the same file
140       via `--video`, or the exact stored `--video-id`. |
141     | `video_id` doesn't match although it's "the same" video | Re-encoding/trimming changes
142       the bytes, hence the MD5. Use the exact file that was processed, or look up the stored id (it's
143       printed during processing). |
144     | `candidates` stage is empty but `final` has data (or vice-versa) | Depends on which
145       segmenter ran. `gemini` writes only `final` (no candidates); `yolo_hybrid`/`tracknet_hybrid`
146       write both. Use `--stage final` for the gemini segmenter, and `--detector <name>` to scope
147       candidates. |
148     | Recall looks low only at high `--iou` | Rallies are found but boundaries are loose ?
149       check `dStart`/`dEnd`. Tune `stitching` padding or the segmentation thresholds rather than the

```

```

detector. |
139 | `ValueError: start (...) >= end (...)` while loading annotations | A bad row in your
CSV. Timestamps must be increasing per rally and use seconds or `HH:MM:SS`. |
140
141 Run `python run_eval.py -h` for a quick reminder of this prerequisite and all flags.
142

```

11. user (2026-06-16T02:25:13.619Z)

```

1 # Experiment harness ? A/B + shadow eval for rally-detection quality (no-regression by
design)
2
3 **Status:** **P1 IMPLEMENTED** (`backend/eval/experiment.py` +
`tests/test_experiment.py`, shipped
4 2026-06-13) ? the `Variant` layer, `compare()` (labeled A/B, LOVO-honest),
`compare_unlabeled()`
5 (SxS on NEW unlabeled footage), and the leaderboard. **P2?P4 remain owner-gated.**
Phases below are
6 sequenced but each is a separate PR.
7
8 > **Companion docs:** the cues this experiments *on* are in
9 > [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); the scoring mechanics are
10 > [EVALUATION.md](EVALUATION.md); the no-regression gate + golden labels +
`baseline.json` come from
11 > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md); the modular-cue architecture is
**ADR-007**
12 > ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md)); the distilled
classifier is
13 > **ADR-004**. The 2-layer mental model is
[HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
14
15 ---
16
17 ## 1. Context ? why this, why now
18
19 PR #112 (merged) added the multi-signal fusion design ? shuttle + person + audio cues,
all **default
20 OFF**. The owner's directive for *how* we evolve quality from here:
21
22 > "Evolve the codebase so we can keep finding new ways to improve quality and
experiment with them ?
23 > or add them as a signal ? without suffering any regressions if the idea doesn't work
out. Rolling
24 > back the code would be too dangerous."
25
26 So **experimentation and deployment must be decoupled.** The contract:
27
28 1. Merge experimental code freely ? it's **gated OFF**, so it cannot change production
behavior.
29 2. Test it **offline against golden labels** (most experiments never touch the served
path at all).
30 3. Promote a variant to default **only on measured lift**, through a CI eval gate.
31 4. **"Rollback" = flip a config flag, never `git revert`.** The proven path is a pinned
variant that is
32 always present.
33
34 **The pleasant surprise (Explore audit, 2026-06-13):** ~80% of this harness already
exists and is
35 simply not composed *as a system*. The missing piece is a thin **variant/experiment
layer + the
36 discipline**, not a platform. S4 is the exact code surface so the next session does not
re-discover it.
37
38 ---
39

```

```

40     ## 2. The thesis ? separate expensive DETECTION from cheap DECISION
41
42     The expensive, GPU/WSL-bound, risky part is **detection**: "what signals exist per
43     frame" (shuttle
44     WASB trajectory, person tracks/boxes, audio hits). The cheap, swappable, pure-Python
45     part is the
46     **decision**: "given those signals, where are the rallies" (windowing + gates + the
47     classifier).
48
49     > **Run detection once per video, persist all per-cue signals + per-candidate features,
50     then re-score
51     > any number of variants OFFLINE ? deterministically, with no GPU and zero production
52     risk.**
53
54     This is not aspirational ? `distill_local.py` and `calibrate_local.py` already do
55     exactly this on
56     stored WASB trajectories. The candidate-segments table (`stats` JSON) is the persistence
57     substrate.
58     ? Most experiments are pure re-scoring of stored data; they never enter the served
59     pipeline.
60     ---
61
62     ## 3. What already exists (the audit ? do NOT rebuild)
63
64     Exact public surface, with `file:line`. **This table is the anti-duplicate-work
65     artifact** ? start here.
66
67     ### 3.1 Scoring (variant-agnostic ? grades any `List[Interval]`)
68     `backend/eval/metrics.py`
69     - `interval_iou(a, b) -> float` (:15) ? temporal IoU of two `(start,end)` intervals.
70     - `match_intervals(preds, gts, iou_threshold=0.5) -> MatchResult` (:48) ? greedy 1-to-1
71     IoU matching.
72     - `score(preds, gts, iou_threshold=0.5, match_result=None) -> Score` (:104) ? P/R/F1,
73     mean IoU, boundary errors; `Score.as_dict()`.
74     - `pr_curve(preds, gts, thresholds=None) -> List[Score]` (:138).
75
76     ### 3.2 A/B + cross-validation primitives (the comparison engine already exists)
77     `backend/eval/calibration.py`
78     - `improvement_report(predict_fn, baseline_config, tuned_config, gts, iou_threshold=0.5)
79     -> dict` (:148) ? **before/after metrics + %?. This IS the A/B delta.**
80     - `leave_one_video_out(videos, configs, metric="f1", iou_threshold=0.5) -> dict` (:113)
81     ? **LOVO: pick on other videos, score on held-out video (honest cross-video).**
82     - `cross_validate(predict_fn, configs, gts, k=5, metric="recall", iou_threshold=0.5) ->
83     dict` (:72) ? within-video k-fold overfit guard.
84     - `select_best(predict_fn, configs, gts, metric="f1", iou_threshold=0.5) -> (Config,
85     float)` (:61).
86     - `evaluate(preds, gts, iou_threshold=0.5) -> Score` (:41); `kfold_indices(n, k)` (:46).
87     - LOVO video dict shape: `{"name": str, "predict_fn": PredictFn, "gts": [Interval]}`.
88
89     ### 3.3 No-regression gate + baselines (the promotion gate already exists)
90     `backend/eval/regression.py`
91     - `regress(db, video_id, ground_truth, stage="final", iou_threshold=0.5, detector=None,
92     tolerance=DEFAULT_TOLERANCE, baseline_path=DEFAULT_BASELINE_PATH) -> RegressionResult` (:104) ?
93     score stored preds vs GT, flag if P **or** R drops > tolerance.
94     - `regress_with_provider(...)` (:160) ? same, fetching GT via the annotation provider.
95     - `save_baseline(video_id, stage, precision, recall, f1, ..., gt_fingerprint=None)`
96     (:80) . `load_baselines(path) -> dict` (:68).
97     - `RegressionResult` (:43): `regressed`, `reasons`, `gt_hash`, baseline P/R,
98     `tolerance`, `has_baseline`; `as_dict()`.
99     - Default baseline file: `eval_baselines/regression_baseline.json` (:33).
100
101     `run_regression.py` ? CI-gatable CLI, `main(argv)` (:52). **Exit codes: 0 ok / 1
102     regression / 2 no-DB / 3 no-baseline.** Flags: `--video-id --tier --annotations
103     --stage{candidates,final} --detector --iou --tolerance --baseline --update-baseline --allow-

```



```

missing-baseline --json`.
84
85     ### 3.4 Pinnable model artifact
86     `backend/eval/classifier.py` ? `LogisticRegression` (pure numpy, L2):
`fit(X,y,feature_names=,class_weight="balanced")` (:40), `predict_proba` (:75),
`predict(threshold=0.5)` (:81), `to_dict`/`from_dict` (:86/:97), `save`/`load` (JSON)
(:107/:111). **A trained model = one JSON file ? a pinnable, hashable variant artifact.**
87
88     ### 3.5 Offline re-scoring substrate (persist once, re-score forever)
89     `backend/infrastructure/database.py`
90     - `add_candidate_segment(video_id, detector, start_time, end_time, chunk_index=None,
confidence=None, stats=None, detection_params=None) -> Optional[int]` (:318) ?
`stats`/`detection_params` stored as JSON; `candidate_segments` table cols incl. `stats`,
`detection_params`, `human_label`, `confirmed_segment_id` (schema :97).
91     - `query_candidate_segments(video_id, detector=None, only_unlabeled=False) ->
List[dict]` (:355) ? returns rows with deserialized `stats`.
92
93     ### 3.6 Label assignment + feature glue
94     `backend/eval/training.py`
95     - `label_candidates(candidate_intervals, gt_intervals, iou_threshold=0.5) -> List[int]`
(:50) ? y-labels by IoU match to golden (same matcher as the evaluator).
96     - `build_training_set(candidate_rows, gt_intervals, iou_threshold=0.5,
feature_fn=extract_yolo_features) -> (X, y, intervals)` (:64) ? pluggable `feature_fn`.
97
98     ### 3.7 Existing LOVO drivers to mirror (not duplicate)
99     - `backend/eval/distill_local.py` ? `run(specs, iou=0.5, threshold=0.5, model_out=None)`
(:98), CLI `python -m backend.eval.distill_local --manifest videos.json --model-out ...`;
`FEATURE_NAMES = [duration, peak_velocity, mean_velocity, active_ratio, net_crossings]` (:40);
already does LOVO over Gemini silver labels.
100     - `backend/eval/calibrate_local.py` ? `run(specs, iou=0.5, metric="f1")` (:77);
`LocalConfig = (velocity_thresh, merge_gap, min_window_duration, min_crossings)`; grid + LOVO.
101     - `backend/eval/calibrate_wasb.py` ? `run(...)` , `load_golden_gts()`;
`backend/eval/gt_loader.py::load_gt_intervals(csv_path, *, strict=True)` is THE canonical golden
reader.
102
103     ### 3.8 Registries + config knobs (variant selection)
104     - `backend/pipeline/segmenters/base.py` ? `SegmenterRegistry` (:35), singleton
`segmenter_registry` (:63); registered (6): `motion`, `gemini`, `yolo_hybrid`,
`tracknet_hybrid`, `wasb_hybrid`, `fusion_hybrid`.
105     - `backend/annotations/base.py` ? `AnnotationProviderRegistry`, singleton
`annotation_registry`; providers `file`, `auto`.
106     - `backend/config/models.py::IndexingConfig` ? `default_segmenter` (:65, default
`"yolo_hybrid"`), `detector_model` (:72), `detector_score_threshold` (:73),
`yolo_velocity_threshold` (:74), `yolo_frame_skip` (:75), `min_segment_duration` (:68),
`dead_time_merge_gap` (:69), `motion_threshold` (:67). The `min_crossings` knob now lives on
`FusionConfig` (default `0` = OFF; also `min_players`, both served Gate A/B thresholds ? see
fusion P2).
107
108     ### 3.9 Confirmed ABSENT (no duplication risk)
109     Explore found **no** existing experiment / variant / shadow / A-B *machinery* ? only a
stray "A/B test"
110     aspiration comment in `backend/pipeline/detectors/base.py` and an "experiment E1" note
in
111     `archives/research/AUDIO_RALLY_DETECTION_PLAN.md`. The `regress()` + baseline path is
the canonical comparison mechanism.
112
113     ---
114
115     ## 4. The variant abstraction (the one thin thing to add)
116
117     A **`Variant` = a declarative recipe**, not a code branch:
118
119     ```
120     Variant = (segmenter_name, IndexingConfig overrides, enabled cues + per-cue params,
classifier model path/hash)

```

```

121     ``
122
123     JSON-serializable; selected via the existing `segmenter_registry` +
`IndexingConfig.default_segmenter`.
124     **Control** = a *pinned, frozen* variant (recipe + model hash) that is always registered
and is the
125     default. Adding a new cue = a new Variant differing by one flag ? the control recipe is
untouched.
126
127     ---
128
129     ## 5. Three experiment modes
130
131     ### 5.1 Offline A/B (primary, label-driven, zero production risk)
132     `compare(videos, variant_A, variant_B)` ? a thin driver composing **existing**
functions:
133     `query_candidate_segments()` (load persisted features) ? re-score each variant ?
`calibration.improvement_report()`
134     + `calibration.leave_one_video_out()` + `metrics.score()`, graded vs golden
(`gt_loader.load_gt_intervals`).
135     Reports per-video **and** LOVO-honest aggregate IoU/P/R/F1 deltas. **This is the A/B
test, and it needs
136     almost no new scoring code ** ? only the glue.
137
138     **Honest reporting (default ON, additive):** `compare(..., honest_stats=True)` also
attaches a "honest"
139     block ? per-metric bootstrap **CIs**, a paired **?F1 CI + exact sign test** (C1), and
**F1@k +
140     segment-count ratio ** (C4) ? *alongside* the bare-mean fields (existing output
unchanged; `honest_stats=False`
141     restores the legacy shape). A bare LOVO mean at n?6 is not promotable on its own; the
lift is real when the
142     ?F1 CI clears 0 **and** the sign test agrees. Wiring of the course corrections in
143     [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13/C1+C4 (helpers:
`backend/eval/eval_stats.py`,
144     `segmentation_metrics.py`); `record_experiment` carries the honest ?F1 into the
leaderboard.
145
146     ### 5.2 Shadow (for cues that touch detection, e.g. the person detector)
147     The new cue runs and **persists its signal** into the candidate
`stats`/`detection_params`, but the
148     *served* decision stays control. Offline you then ask "what would variant B have
scored?" via §5.1 ?
149     B never affects output until it wins. This is how a detection-touching change (not just
a re-scoring
150     change) earns its way in without risk.
151
152     ### 5.3 Promotion gate (no-regression, CI-enforced)
153     `regression.regress()` + `baseline.json` + `run_regression.py` exit codes. A variant
becomes the new
154     default **only if** it doesn't regress the golden set beyond `tolerance` (exit 1
blocks). Control is
155     pinned ? **rollback = flip `default_segmenter`/variant config, never a code revert.**
156
157     ---
158
159     ## 6. No-regression guarantees (the heart of the owner's ask)
160
161     - **New cues/signals default OFF** ? per-cue enable flag in `IndexingConfig`; merged
code can't change
162     behavior until explicitly enabled.
163     - **Control variant pinned** ? frozen recipe + classifier model hash, always registered
& default.
164     - **Promotion only via the eval gate** ? `run_regression.py` in CI (exit 1 blocks); no
silent default change.

```

```

165     - **Experiments are records** ? variant-spec hash ? per-video + LOVO scores + label-set
hash + timestamp,
166     persisted to a small JSON **experiment leaderboard** (generalizes `baseline.json`).
Answers "which
167     signals ever helped, on which video slices," and makes every result reproducible.
168     - **Decoupled events** ? "merge an experiment" and "change the default" are separate,
independently
169     revertible actions.
170
171     ---
172
173     ## 7. What to build later (gap list ? thin, additive, owner-gated)
174
175     1. ~~~**`backend/eval/experiment.py`** ? a `Variant` dataclass + `compare(videos, a, b)`
composing the §3
176     functions (no new scoring math).~~ **DONE (2026-06-13).** Shipped `Variant` (+ pinned
`CONTROL`),
177     `compare` (labeled A/B with LOVO-honest learned-variant training, mirroring
`distill_local`),
178     `compare_unlabeled` (the **SxS-on-new-videos** mode ? agreed / A-only / B-only via
`match_intervals`,
179     no GT), `record_experiment` (? `eval_baselines/experiment_leaderboard.json`), and
`load_video_candidates`
180     (the `query_candidate_segments` bridge). JSON variant recipes first as recommended; a
`VariantRegistry`
181     only if the count grows. 19 tests incl. the no-regression invariant.
182     2. **Persist per-cue features** into candidate `stats` (person: `players_in_court`,
`two_sided_motion`,
183     `in_court_ratio`, `shuttle_player_colocation`; shuttle-state: `net_crossings`,
`direction_reversals`;
184     audio: hit times) so any future variant re-scores offline ? ties to
`MULTI_SIGNAL_FUSION_PLAN.md` §3.2/§4.
185     3. **Per-cue enable flags** in `IndexingConfig` (default `False`); `default_segmenter`
already exists.
186     Add the `min_crossings` knob (fusion P2) here too.
187     4. **CI promotion lane** using `run_regression.py` against the golden `baseline.json`;
write the
188     experiment-leaderboard JSON.
189
190     ---
191
192     ## 8. Phasing (owner-gated; ONE PR per slice, off `master`)
193
194     - **P0 ? this doc**, indexed in `docs/README.md`, cross-linked from
`MULTI_SIGNAL_FUSION_PLAN.md` and
195     `GOLDEN_SET_VENDORING.md`. *(this deliverable; docs-only, no code.)*
196     - **P1 ? `experiment.py` `compare()`** ? **DONE (2026-06-13):** thin wrapper over
`calibration` +
197     `metrics` + `query_candidate_segments`; pure-logic, fully tested in this dev container
(`compare`,
198     `compare_unlabeled`, `record_experiment`, the no-regression invariant).
`compare_unlabeled` adds the
199     SxS read on **new unlabeled footage** the owner asked for (run control + candidate on
a fresh video,
200     diff where they agree/diverge ? two reels via the existing `VideoCompiler` is the GPU-
side follow-on).
201     - **P2 ? per-cue enable flags + feature persistence** into candidate `stats` (default
OFF) ? couples
202     with fusion P1/P2.
203     - **P3 ? CI promotion gate**: `run_regression.py` lane on the golden `baseline.json`;
experiment-leaderboard JSON.
204
205     ---
206
207     ## 9. Scope guardrail

```

```

208
209     **Single-tenant / local-first** (per `PLATFORM_ARCHITECTURE.md`): a JSON experiment
registry + the
210     existing harness ? explicitly **NOT** MLflow / W&B / a service. Keep it the size of
`baseline.json`.
211
212     ---
213
214     ## 10. Verification
215
216     - **Offline, runnable in this dev container (no GPU/torch needed for the glue):** unit
test that
217         `compare()` is deterministic on synthetic candidate rows; a **no-regression invariant
test** asserting
218         "all new-cue flags OFF ? predictions byte-identical to control"; a CI-gate test
asserting
219         `run_regression.py` exit codes (0 ok / 1 regress / 3 no-baseline).
220     - **Machine-side (owner, GPU/WSL):** persist candidates once on a real clip, run
`compare(control,
221     fusion)` LOVO, confirm the reported lift matches a manual `metrics.score()` spot-
check.
222     - **Docs (P0):** links resolve; indexed in `docs/README.md`; cross-linked from
223         `MULTI_SIGNAL_FUSION_PLAN.md` and `GOLDEN_SET_VENDORING.md`.
224     </content>
225

```

12. user (2026-06-16T02:25:13.863Z)

```

1     # Ablation layer (Phase 0) ? per-signal marginal contribution
2
3     A *systematic, repeatable* version of the earlier one-off ablation:
4     `backend/eval/ablation.py` runs it and it EXCLUDES one signal, measures whether quality
dropped,
5     and reports each signal's **marginal ?F1** (full set vs full-minus-that-signal) with a
bootstrap
6     95% CI + paired sign test + the C4 over-segmentation ratio. The broader design (the
ledger time
7     series, interaction handling, per-user ablation, the phased plan) is not yet split into
its own
8     doc; this note covers only the shipped Phase-0 runner.
9
10    ## What it is (and is not)
11
12    **A pure driver.** No new scoring/stats math ? every number defers to the shipped
13    `experiment.compare` + `eval_stats` engine, exactly as `compare` was a driver over
14    `metrics`/`calibration`. For a signal `s`:
15
16    ```
17    ?(s) = score(full set S) ? score(S \ {s})          # leave-one-out drop, ?F1 primary
18    ```
19
20    is `experiment.compare(videos, variant_a=without_s, variant_b=full)` ? A is the ablated
(smaller)
21    variant, B is full, so the reported **B?A** ?F1 IS ?(s): a per-video-paired ?F1 with
bootstrap CI
22    + exact sign test, LOVO-honest for learned variants.
23
24    A signal is either a **feature column** (person/audio/shuttle columns the learned scorer
weights,
25    ablated by dropping the column) or a **gate knob** (Gate-A's `min_crossings` threshold,
ablated by
26    zeroing the knob). The runner handles both uniformly.
27
28    ## Two granularities
29

```

```

30 - **LOSO** (`--granularity signal`) ? leave one column-set out at a time (fine-grained).
31 - **LOGO** (`--granularity group`) ? leave a whole producer's columns out at once (the
owner's
32 "is this old SIGNAL still relevant?" question at cue granularity). Default.
33
34 ## The honest small-N contract
35
36 Inherited from `eval_stats` / `per_user_gate` (no new statistics):
37
38 - **Never a bare mean** ? every row ships a CI + the exact sign test.
39 - A signal **earns its keep** only when removal drops F1 with the ?F1 **CI excluding 0
AND the
40 sign test agreeing** (the `per_user_gate` bar, in reverse).
41 - **Structural guards are exempt from auto-demotion.** Gate-A (the held-shuttle false-
positive
42 killer) is marked `structural`; it is reported honestly but verdict is always
43 `structural-protected`, never flagged dead weight on "no measured lift" ? its value is
in
44 failure modes the golden corpus may not contain.
45 - The verdict vocabulary is closed and never over-claims: `clears-bar` / `suggestive` /
`inert`
46 / `structural-protected`. One wash at n=6 is `inert` (a candidate), never
"irrelevant".
47
48 ## Run it
49
50 ```bash
51 # LOGO contribution table over the learned full set (shuttle+gate_a+person+audio):
52 python -m backend.eval.ablation --features-dir output [--granularity signal] \
53     [--pdf ablation.pdf] [--json ablation.json]
54
55 # Reproduce the hand-measured Gate-A result (the framework's litmus):
56 python -m backend.eval.ablation --features-dir output --litmus
57 ```
58
59 The corpus is the 6 persisted golden feature JSONs (`output/*_golden_features.json`,
written by
60 `fusion_golden.py` ? persist once on the GPU, re-score forever offline). `--pdf` writes
a clean
61 reportlab report (table of marginal contribution + CI + sign test + a verdict column).
62
63 ## Litmus (regression guard on the math)
64
65 `tests/test_ablation.py::test_litmus_reproduces_gate_a` asserts that, on the 6 real
golden JSONs,
66 the runner reproduces the recorded `QUALITY_ITERATIONS` Gate-A result within tolerance:
67
68 | metric | recorded (2026-06-14) | reproduced |
69 |---|---|---|
70 | marginal ?F1 | +0.074 | +0.074 |
71 | sign test | 6W/0L, p=0.031 | 6W/0L, p=0.031 |
72 | 95% CI | [+0.042, +0.104] | [+0.042, +0.105] |
73 | over-seg ratio | 1.68 ? 1.01 | 1.68 ? 1.01 |
74
75 The test skips cleanly when the golden substrate is absent (CI) and runs when present
(owner box).
76 The rest of the suite runs GPU-free on deterministic synthetic candidate rows.
77

```

13. assistant (2026-06-16T02:25:16.908Z)

Now the last two: the Roora labeling schema/versioning and the quality-iterations empirical backbone. These directly answer questions 2 (per-video artifacts in repo) and 3 (schema/versioning).

14. user (2026-06-16T02:25:17.821Z)

```
1      # Roora Annotation Guidelines ? Rally Golden Set
2
3      **Version: v8 (2026-06-13).** Supersedes the older Drive copies (v3/v5/v6/v7) **and**
the internal
4      "DRAFT v2" label ? those numbers never matched each other; **v8 is the single label to
use with
5      Roora going forward.** Content is otherwise the finalized 2026-06-12 guideline plus the
6      fusion-driven additions marked **`? ADDED (fusion)`** below.
7
8      > **?? Canonical home / provenance.** **This in-repo file is the sole canonical copy of
the guideline**
9      > ? it lives here because the multi-signal-fusion features that motivate the new fields
(court
10     > calibration, singles/doubles) originate in *this* repo (see
11     > [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md) and
12     > [GOLDEN_SET_VENDORING.md](GOLDEN_SET_VENDORING.md), ADR-006). The annotation
deliverable still imports
13     > into the sibling **`sports-data-collector`** repo via `python -m src.cli.curate
import-local`; if a
14     > mirror is ever staged there for the collector's intake docs, it must be regenerated
from this file
15     > (and the `? ADDED (fusion)` callouts are exactly the deltas to port).
16     >
17     > **`TODO(avidullu)` markers are preserved** ? they are owner-only items (Drive links,
intake
18     > manifest, NDA counsel review) and must be filled before sending to Roora. Search for
`TODO(avidullu)`.
19
20     Brief (vendor spec) + Illustrated Guide (rater how-to), in one doc.
21
22     ---
23
24     # PART A ? Roora Annotation Brief (vendor-facing spec)
25
26     **Project:** racket-sport rally segmentation golden / regression dataset.
27     **Sports in scope:** Badminton, Pickleball, Table Tennis (each in singles AND doubles).
28
29     This brief is the vendor-facing spec. The annotation deliverable is a per-rally CSV that
imports
30     directly via `python -m src.cli.curate import-local`.
31
32     ## 0. What we need in one sentence
33
34     For each provided match video, produce ONE CSV file listing EVERY rally with its start
time, end
35     time, number of shots, and how it ended ? EXCLUDING all dead time.
36
37     ## 1. Deliverable format (canonical)
38
39     One CSV per video (UTF-8), named exactly `.csv` (we supply the `video_id`).
One row per
40     rally, chronological, header row required.
41
42     **Columns:**
43
44     | Column | Type / values | Meaning |
45     |---|---|---|
46     | `rally_number` | integer, 1-based, chronological ? no gaps, no repeats | rally index |
47     | `start_mmss` | `m:ss.mmm` (e.g. `1:12.40`) | rater-entered serve-contact time |
48     | `end_mmss` | `m:ss.mmm` | rater-entered ball/shuttle-dead time |
49     | `start_time` | decimal **seconds** (e.g. `72.40`) ? **AUTO-FILLED** from `start_mmss`
| machine time |
50     | `end_time` | decimal **seconds** ? AUTO-FILLED from `end_mmss` | machine time |
```

```

51 | `shots_count` | integer | total racket/paddle/bat contacts (serve = shot 1) |
52 | `ending_reason` | `winner` \ | `forced_error` \ | `unforced_error` \ | `service_fault` \ |
`let` \ | `other` | WHY it ended |
53 | `last_shot_outcome` | `in` \ | `net` \ | `out` \ | `n_a` | WHERE the last shot landed
(`n_a` for service_fault/let) |
54 | `score_before` | optional, "A-B" ? blank if not confidently readable | score before
|
55 | `score_after` | optional, "A-B" | score after |
56 | `notes` | optional free text ? **REQUIRED when `ending_reason` = other** | edge-case
note |
57
58 **Times (mm:ss ? decimal seconds):** raters type only `start_mmss` / `end_mmss`. The
template Google
59 Sheet fills `start_time` / `end_time` with this formula (shown for `start_time`, row 2):
60
61 ```
62 =IF(B2="", "", IFERROR(VALUE(LEFT(B2,FIND(":",B2)-1))*60 +
VALUE(MID(B2,FIND(":",B2)+1,20)), VALUE(B2)))
63 ```
64
65 On "Download as CSV" the computed decimal seconds export. (If your tool already shows
decimal
66 seconds, type those into `start_time` / `end_time` and leave the mm:ss columns blank.)
67
68 **Example (badminton):**
69
70 ```
71 rally_number,start_mmss,end_mmss,start_time,end_time,shots_count,ending_reason,last_shot
_outcome,score_before,score_after,notes
72 1,0:12.50,0:25.00,12.50,25.00,9,winner,in,0-0,1-0,
73 2,0:45.00,0:47.30,45.00,47.30,1,service_fault,n_a,1-0,1-1,serve into net
74 3,1:01.20,1:18.85,61.20,78.85,14,unforced_error,out,1-1,2-1,final clear sailed long (no
pressure)
75 ```
76
77 **Critical format rules:**
78 - `start_time` / `end_time` that reach us MUST be decimal seconds (the formula
guarantees this).
79 - Times are relative to the **START OF THE FILE WE PROVIDE**. Do NOT trim, clip, or re-
encode the
80 video ? our pipeline identifies a video by its file checksum. (The file we share may
be an
81 **annotation-optimized copy** of the original ? lighter and faster to frame-step, but
82 frame-for-frame identical. Treat it exactly like the original: annotate it as-is.)
83 - `end_time` > `start_time` for every row.
84 - No overlaps: `end_time[N]` <= `start_time[N+1]`.
85 - Everything between one rally's `end_time` and the next rally's `start_time` is DEAD
TIME and gets
86 NO row.
87
88 ## 2. The three non-negotiable rules (full detail + pictures in the Guide)
89
90 1. **IGNORE ALL DEAD TIME.** Only label active play.
91 2. **INCLUDE EVERY RALLY ? EXCLUDE NONE.** Every served point, including service faults,
1?? shot
92 rallies, and lets/replays. A missed rally is the worst error in this project.
93 - **EXCEPTION:** warm-up / knock-up is NOT a rally ? pre-match and between-games
warm-up looks like
94 rallying but isn't scored play, so don't label it. Tell it apart by: no real serve;
cooperative
95 (not competitive) hitting; the score not progressing; often several shuttles/balls
in use (full
96 how-to in the Guide, Part 2). When unsure if the match has started, flag it.
97 - **SECOND EXCEPTION:** a rally already in progress when the file starts (or still in
progress when

```

```

98         it ends) gets NO row ? its true start (or end) is outside the footage and must
never be guessed
99         or extrapolated. Note it in the delivery instead.
100     3. **A SHOT = ONE CONTACT** with the racket/paddle/bat. Serve = shot 1. Bounces and net-
clips are not shots.
101
102     ## 3. Scope & phasing
103
104     - **Phase A ? Pilot (this engagement):** 3 videos (1 badminton + 1 pickleball + 1 table
tennis), each
105     < 3 min with ~10?12 rallies. This is the paid calibration round; we review jointly and
lock the
106     Guide before scale-up.
107     - **Phase B ? Scale (after a successful pilot):** ~12?15 videos PER SPORT (~36?45
total), each ~30 min.
108     Same format, rules, and template.
109
110     Doubles are included across the program. Rally rules are identical to singles; the
badminton- and
111     pickleball-specific doubles notes are in the Guide.
112
113     ## 4. Quality bar (acceptance criteria)
114
115     Automated validation (no overlaps, positive durations, schema-valid) runs on every file,
then a ~20%
116     human spot-check. A file is returned for rework (in scope) if it misses:
117
118     - **Completeness:** every rally present; zero missed rallies; zero dead-time rows.
119     - **Boundaries:** `start_time` / `end_time` within  $\pm 0.3$  s of true serve-contact /
landing.
120     - ?? Boundaries that all sit on a frame-sampling grid (e.g. every 30th frame)
STRUCTURALLY FAIL this
121     bar and are detected automatically ? the whole file is returned for rework. See §7
for the
122     required method.
123     - **`shots_count`:** exact for rallies ? 15 shots;  $\pm 1$  allowed for longer/faster rallies.
124     - **`ending_reason`:** one of the six allowed values, per the Guide's decision tree.
125     - **Format:** decimal seconds present, valid columns, non-overlapping, chronological.
126
127     > ?? ADDED (fusion) ? court calibration as a day-1 quality lever. The owner's
pipeline produced a
128     > bad highlight where a held shuttle was mislabeled as a rally. The boundary definition
($ Part 3:
129     > start = serve contact, not the toss/hold) already excludes that ? so faithful
boundaries are
130     > themselves a quality checkpoint. The one NEW field we need (next section) is court
calibration,
131     > which lets the detector reason about court occupancy and net-crossings.
132
133     ## A. Court calibration & intake manifest    ? ADDED (fusion)
134
135     > ?? ADDED (fusion). *Why: the multi-signal detector
136     > ([MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md)) needs to know where the
court is and
137     > where the net is to (a) ignore people outside the court, (b) require ?2 players on
court for a
138     > rally, and (c) count shuttle net-crossings. This is a one-time, per-video
annotation ? cheap,
139     > high-leverage, and required in deliverables from day 1.
140
141     Per-video, recorded once in the intake manifest (NOT in the per-rally CSV):
142
143     | Manifest column | Type | Meaning |
144     |---|---|---|
145     | `video_id` | string | (existing) we supply it |

```



```

146 | `fps` | number | (existing) we supply it |
147 | `court_corners` | 4 points `(x,y)` in pixels, clockwise from the near-left corner |
the playable court rectangle |
148 | `net_line` | 2 endpoints `(x,y)` in pixels | the net, left post ? right post |
149 | `singles_or_doubles` | `singles` \ | `doubles` | sets the expected on-court player
count (2 vs 4) |
150
151 - Pick one clear frame** with the empty/visible court and click the 4 corners + 2 net
endpoints.
152 - If the camera does not move for the whole video (the normal case), one calibration
covers the file.
153 If the camera is moved mid-file, flag it ? we'll split the file.
154 - Worked example (badminton, 1920x1080):
`court_corners=[[360,980),(1560,980),(1330,300),(590,300)]`,
155 `net_line=[[470,640),(1450,640)]`, `singles_or_doubles=singles`.
156
157 *(After the first batch we may add more per-shot fields ? see the §B appendix ? and
renegotiate price.)*
158
159 ## 5. Per-sport quick reference (full version + examples in the Guide)
160
161 - Badminton (singles & doubles)** ? start: racket?shuttle serve contact . end: shuttle
floors /
162 fault . shot: each racket?shuttle contact.
163 - Pickleball (singles & doubles)** ? start: paddle?ball serve contact . end: double
bounce / out /
164 net / fault . shot: each paddle?ball contact (floor bounces are NOT shots; two-bounce
rule is normal play).
165 - Table tennis** ? start: racket?ball serve strike (not the toss) . end: failed legal
return . shot:
166 each racket?ball contact (table bounces are NOT shots). Net-cord serve = let.
167
168 ## 6. `ending_reason` values (decision tree in the Guide)
169
170 This vocabulary attributes WHY the point ended; it is shared across all net-separated
sports
171 (badminton, pickleball, table tennis, tennis). Record WHERE the last shot physically
landed in the
172 separate `last_shot_outcome` column (`in` | `net` | `out` | `n_a`) ? e.g. a forced error
that flew
173 long is `ending_reason=forced_error, last_shot_outcome=out`.
174
175 - `winner` ? last shot landed IN and the opponent made no legal contact (a clean
winner).
176 - `forced_error` ? the losing player erred (netted / out / missed) UNDER PRESSURE ?
stretched,
177 wrong-footed, or rushed by the opponent's strong shot.
178 - `unforced_error` ? the losing player erred on a ROUTINE, UNPRESSURED ball they had
time and position
179 to make.
180 - `service_fault` ? the serve itself failed / was illegal.
181 - `let` ? the point was replayed.
182 - `other` ? anything else / cannot tell. MUST add a note.
183
184 ## 7. Capture method (required for boundary frames)
185
186 Use any tool that shows current time (`mm:ss.mmm` or seconds, or frame + known fps) and
can step one
187 frame forward/back: a frame-steppable player (VLC frame-step) or a timeline tool (Label
Studio /
188 CVAT). The deliverable is the CSV in the schema above ? or, if you work in CVAT, the
CVAT export as-is
189 (we recompute each rally's time from its box frame range; seconds = `frame_number /
fps`, fps provided
190 per video).

```

```

191
192     **Boundary placement is the non-negotiable part:**
193     - Set up the CVAT task at **FRAME STEP 1** on the file we provide (it is encoded to make
this fast ?
194     frame-stepping and back-scrubbing stay snappy).
195     - The FIRST box of every rally goes on the EXACT serve-contact frame; the LAST box on
the EXACT
196     shuttle/ball-dead frame. Frame-step to find them.
197     - Boxes IN BETWEEN may stay sparse (e.g. every 30th frame) ? only the two boundary
frames must be exact.
198     - Self-check before submitting: every start/end within  $\pm 0.3$  s of the true frame (Guide).
199
200     ## 8. Logistics & contacts
201
202     - **TODO(avidullu):** paste the Google Drive folder link where source videos are shared
with Roora.
203     - Per-video fps + `video_id` (+ the calibration columns from $A): provided in the intake
manifest
204     (template `INTAKE_MANIFEST_TEMPLATE.csv`). **TODO(avidullu):** fill the manifest (one
row per video)
205     and share its link.
206     - **TODO(avidullu):** paste the Drive folder link where Roora uploads completed
`<video_id>.csv` files.
207     - Blank annotation template Sheet (mm:ss ? seconds formula from $1 pre-filled):
208     **TODO(avidullu):** create it and paste the link.
209     - **TODO(avidullu):** add the escalation channel (email / WhatsApp group) for edge-case
questions.
210     - NDA before any footage is shared ? template `NDA_TEMPLATE.md`. **TODO(avidullu):**
have counsel
211     review, fill the blanks, and have Roora sign.
212     - Roora contact on file: info@rooragrp.com (HSR Layout, Bangalore).
213
214     > **? ADDED (fusion) ? data-handling guardrails (put in the brief / contract):**
215     > - **Exclude minors entirely.** If footage contains children, that video does **not**
enter the
216     > labeled pool ? do not annotate it; flag it back to us.
217     > - **No biometric / identity labeling.** Refer to players only by **per-video
pseudonyms** (e.g.
218     > "near side" / "far side") ? never a real name, and never a cross-video person ID. We
do **not**
219     > collect gait or biometric annotations.
220     > - **No pose / skeleton / keypoint annotation** is requested.
221     > - **Work-for-hire IP assignment** (atop the NDA): the labels Roora produces are
assigned to us as
222     > owned work product.
223
224     ## 9. Pricing request (please quote)
225
226     1. Price per rally and per minute of footage.
227     2. Pilot price (3 short videos, ~30?36 rallies) as a paid calibration round.
228     3. Indicative scale price for ~36?45 longer videos.
229     4. Turnaround for the pilot and throughput (rallies/day) at scale.
230     5. QA process and rework policy.
231     6. Whether annotators have racket-sport familiarity (helpful, not required).
232     7. Confirmation you can deliver the CSV schema above **(incl. the $A intake-manifest
calibration columns).**
233
234     ---
235
236     # PART B ? Illustrated Annotation Guide (rater how-to)
237
238     **Sports:** Badminton, Pickleball, Table Tennis (singles AND doubles).
239
240     > **Illustrations:** boxes marked `[SCREENSHOT: ...]` are placeholders.
241     > **TODO(avidullu):** replace every `[SCREENSHOT: ...]` placeholder with a real frame

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```

grab from the
242     > pilot videos before sending this guide to Roora.
243
244     ## Part 1 ? What you are producing
245
246     One CSV per video (from the template Sheet), one row per rally:
247     `rally_number, start_mmss, end_mmss, start_time, end_time, shots_count, ending_reason,
last_shot_outcome, score_before, score_after, notes`.
248
249     You type: `rally_number`, `start_mmss`, `end_mmss`, `shots_count`, `ending_reason`,
`last_shot_outcome`
250     (+ optional score/notes). The `start_time` / `end_time` decimal-second columns FILL
THEMSELVES via a
251     formula already in the template.
252
253     For every rally you record:
254     - When it started (serve contact) ? `start_mmss` (`m:ss.mmm`, e.g. `1:12.40`)
255     - When it ended (shuttle/ball dead) ? `end_mmss`
256     - How many shots happened ? `shots_count`
257     - Why it ended ? `ending_reason` · Where the last shot landed ? `last_shot_outcome`
258
259     Everything else (gaps between rallies) you IGNORE.
260
261     ## Part 2 ? The three golden rules
262
263     **RULE 1: IGNORE ALL DEAD TIME.** A rally is CONTINUOUS active play. The moment play
stops, the rally
264     is over. The time until the next serve is DEAD TIME and is never labeled.
265
266     ...
267     DEAD          RALLY 1          DEAD          RALLY 2          DEAD
268     (ignore) |=====| (ignore) |=====| (ignore)
269             ^               ^               ^               ^
270             serve          shuttle          serve          shuttle
271             contact        lands            contact        lands
272             =start_1       =end_1           =start_2       =end_2
273     ...
274
275     Dead time includes: walking back to serve, picking up the shuttle/ball, wiping sweat,
bouncing the
276     ball before serving, the score being announced, replays/slow-mo, crowd or coach shots,
changeovers,
277     towel/medical timeouts, and PRE-MATCH / BETWEEN-GAMES WARM-UP (knock-up) ? see the Rule
2 caveat below.
278
279     **RULE 2: INCLUDE EVERY RALLY ? EVEN THE UGLY ONES.** Every served point gets a row,
including:
280     - Service faults (serve into net/out/illegal) ? usually `shots_count = 1`,
`ending_reason = service_fault`.
281     - Very short rallies (serve + one error).
282     - Lets / replays ? `ending_reason = let`; the replayed point is its OWN next row.
283
284     A missed rally is the worst error in this project ? the model must learn that short and
failed rallies
285     are still rallies.
286
287     > ?? **BUT WARM-UP / KNOCK-UP IS NOT A RALLY** ? a common trap, because Rule 2 says
"include
288     > everything." Pre-match and between-games warm-up LOOKS like rallying but is not scored
play; do not
289     > label it. How to tell it apart:
290     > - **No real serve.** Real play starts with a proper serve from the service position
(diagonal,
291     > behind the service line). Warm-up has players feeding cooperatively from mid-court
with no serve.

```

```

292 > - **Cooperative, not competitive.** Gentle hits straight to each other; nobody is
trying to win the
293 > point (no smashes-to-win, no scrambling/diving).
294 > - **Score isn't progressing.** No umpire / scoreboard not running / players casually
retrieve and
295 > re-feed; the score doesn't change. Often several shuttles/balls in use at once.
296 > - **Per sport:** badminton & pickleball ? real play starts with the underhand serve
from the service
297 > court / behind the baseline; table tennis ? real play starts with the tossed, two-
bounce serve.
298 >
299 > When unsure: find the first proper serve where the score starts to progress ? labeling
begins there.
300 > If you genuinely can't tell whether the match has started, flag the clip to us ? don't
guess.
301
302 **RULE 3: A SHOT = ONE CONTACT WITH THE RACKET / PADDLE / BAT.** Count every time the
shuttle/ball
303 touches a racket, paddle, or bat. The serve is shot 1.
304 - Count: any racket/paddle/bat contact (serve, return, smash, block?).
305 - Don't count: floor bounces, table bounces, net clips, the toss before a serve.
306
307 5-shot example: serve(1) ? return(2) ? hit(3) ? hit(4) ? winner(5) ? `shots_count = 5`,
308 `ending_reason = winner`.
309
310 ## Part 3 ? Where exactly does a rally start and end?
311
312 **START (`start_mmss`):** the exact frame the server's racket/paddle/bat CONTACTS the
shuttle/ball.
313 **Not the toss, not the wind-up ? the contact frame.** `[SCREENSHOT: serve-contact
frame]`
314
315 > **? NOTE (fusion):** this is the field that fixes the "held-shuttle" false positive ?
a player
316 > holding/flicking the shuttle before serving is **dead time**, so it never gets a row.
Faithful
317 > serve-contact boundaries are exactly what the detector is graded against.
318
319 **END (`end_mmss`):** the exact frame the shuttle/ball becomes DEAD:
320 - Badminton: shuttle touches the floor, or play stops on a fault.
321 - Pickleball: the second floor bounce, or the frame the ball hits the net / lands out.
322 - Table tennis: the second bounce on one side, a non-serve net hit, or off the table.
323
324 `[SCREENSHOT: end frame]`
325
326 If occluded/off-screen, pick the closest clear frame and add a note. **Boundary
tolerance:  $\pm 0.3$  s
327 ( $\sim \pm 9$  frames at 30 fps).**
328
329 ## Part 4 ? `ending_reason`: how to choose
330
331 Pick exactly ONE value for how the rally ended. Stop at the first match:
332 1. Did the SERVE itself fail / was illegal (and ended the point)? ? `service_fault`
333 2. Was the point REPLAYED (let / interruption)? ? `let`
334 3. Did the final shot land IN and the opponent make NO legal contact? ? `winner` (a
clean winner)
335 4. Did the losing player make an error (netted / out / missed) UNDER PRESSURE ?
stretched,
336 wrong-footed, or rushed by a strong shot? ? `forced_error`
337 5. Did the losing player make an error on a ROUTINE, UNPRESSURED ball they had the time
and position
338 to make? ? `unforced_error`
339 6. None of the above / cannot tell ? `other` (add a note)
340
341 **FORCED vs UNFORCED ? the one judgment call.** Decide using the opponent's PRECEDING

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shot:
342     - `forced_error` = the opponent's shot CAUSED the miss (pace, placement, deception; the
player barely
343     reached it or was pulled out of position).
344     - `unforced_error` = a clean, makeable ball the player missed on their own.
345     - When genuinely unsure between the two, choose `unforced_error` and add a brief note.
346
347     Notes:
348     - A clean winner means NO legal contact by the opponent. If they got a racket/paddle/bat
on it but
349     netted or hit out, that is a `forced_error` / `unforced_error` (their shot), not a
winner.
350     - A serve into the net is `service_fault` (rule 1 beats everything).
351
352     **THEN SET `last_shot_outcome`** ? where the last shot physically went. This is a
SEPARATE,
353     observational column (no judgment): just look at the last shot.
354     - `in` ? landed in (winners; also a put-away the opponent couldn't reach).
355     - `net` ? went into the net.
356     - `out` ? landed outside the court / past the lines.
357     - `n_a` ? not applicable: use for `service_fault` and `let`.
358
359     So each rally gets both: WHY (`ending_reason`) and WHERE (`last_shot_outcome`) ? e.g. a
smash winner =
360     `winner, in`; a clear forced long = `forced_error, out`; a routine ball netted =
`unforced_error, net`.
361
362     ## Part 5 ? Per-sport detail
363
364     ### Badminton
365     - **Serve:** underhand, below the waist, diagonal. Start = racket?shuttle contact.
366     - **Rally end:** shuttle hits the floor, or play stops on a fault.
367     - **Shots:** every racket?shuttle contact; serve = shot 1.
368     - **Service faults** (1-shot rally, `service_fault`): serve into net, serve outside
service court,
369     contact above waist, foot fault.
370
371     Worked example: `[SCREENSHOT: badminton serve]`
372     - Rally 4: serve `1:28.30`, long rally, smash winner, shuttle lands `1:44.10`, 12
contacts.
373     `4, 1:28.30, 1:44.10, (auto), (auto), 12, winner, in, 3-5, 4-5,`
374     - Rally 5: serve `1:57.00` straight into the net.
375     `5, 1:57.00, 1:57.90, (auto), (auto), 1, service_fault, n_a, 4-5, 4-6,`
376
377     **Badminton ? doubles (2 per side).** All rally rules are IDENTICAL to singles:
378     - start = serve contact; end = shuttle down / fault; a SHOT = any racket?shuttle contact
by EITHER
379     player on a side.
380     - `shots_count` = TOTAL contacts across both teams in the rally. Do not track which
player hit ? just
381     count every contact.
382     - Net exchanges (drives, flat pushes) come fast ? frame-step so you neither miss nor
double-count.
383     - Only the designated receiver returns the serve; if the wrong player returns, the rally
ends on a
384     fault ? still a row (`service_fault` if it's a serve/receiving-position fault, else
`other` + note).
385     - Two teammates may NOT both hit the shuttle in succession (double-hit). Two contacts on
the SAME side
386     back-to-back = a fault ending the rally ? count both contacts, then the rally ends.
387
388     ### Pickleball
389     - **Serve:** underhand, paddle below waist, hit diagonally; must clear the non-volley
zone (the
390     "kitchen"). Start = paddle?ball contact.

```

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391 - **Two-bounce rule:** the serve must bounce once in the receiver's court and the return
must bounce
392 too, before either side may volley. These bounces are NORMAL play, not rally ends, and
bounces are
393 NOT shots.
394 - **Rally end:** ball bounces twice, lands out, hits the net (fails to cross), or a
fault (e.g.
395 volleying in the kitchen).
396 - **Shots:** every paddle?ball contact only; serve = shot 1; floor bounces are not
shots.
397 - **Service faults** (`service_fault`): serve into net, out/long, short in the kitchen,
foot fault.
398
399 Worked example: `[SCREENSHOT: pickleball serve + kitchen line]`
400 - Rally 2: serve `0:33.10`; serve bounces, return bounces, a few volleys; receiver nets
an easy one at
401 `0:41.85`; 6 paddle contacts.
402 `2, 0:33.10, 0:41.85, (auto), (auto), 6, unforced_error, net, 1-0, 2-0, easy ball
netted`
403
404 **Pickleball ? doubles (the standard format, 2 per side).** All rally rules IDENTICAL to
singles:
405 - start = serve contact; a SHOT = any paddle?ball contact by EITHER teammate; floor
bounces still not
406 shots; two-bounce rule still applies.
407 - `shots_count` = TOTAL paddle contacts across both teams; don't track which player.
408 - Serving sequence (FYI only ? you do NOT annotate it): both partners serve before the
serve passes to
409 the opponents, except the very first service turn of the game. The called score has
three numbers.
410 Ignore for annotation; just mark boundaries / shots / ending.
411 - If both teammates contact the ball in succession, that's a fault ? rally ends; count
the contacts,
412 set the ending per the fault.
413
414 ### Table tennis
415 - **Serve:** ball tossed up from an open palm, then struck so it bounces once on each
side. Start = the
416 racket?ball strike (not the toss).
417 - **Rally end:** a player fails a legal return ? ball double-bounces on their side, goes
off the table,
418 or is hit into the net on a non-serve shot.
419 - **Shots:** every racket?ball contact only; serve = shot 1; table bounces are not
shots.
420 - **Let:** a serve that clips the net but is otherwise legal ? the point is replayed.
421 `ending_reason = let`, and the replay is its own next row.
422 - **Service faults** (`service_fault`): missed/illegal toss, serve into net, serve off
the table.
423 - **Speed warning:** contacts can be < 0.2 s apart ? step frame-by-frame to count shots.
424 - **Doubles** (if a clip appears): partners must hit in STRICT ALTERNATION and serve
diagonally; a
425 missed alternation is a fault that ends the rally. Same rule ? count every racket?ball
contact as a
426 shot; `shots_count` is the team total.
427
428 Worked example: `[SCREENSHOT: table-tennis serve contact]`
429 - Rally 9: serve `4:10.40`; fast 7-contact rally; forehand winner, opponent no contact;
dead `4:14.05`.
430 `9, 4:10.40, 4:14.05, (auto), (auto), 7, winner, in, 6-9, 6-10,`
431 - Rally 10: serve `4:22.10` clips the net (otherwise good) ? let.
432 `10, 4:22.10, 4:23.00, (auto), (auto), 1, let, n_a, 6-10, 6-10, net-cord serve,
replayed`
433
434 ## Part 6 ? Edge cases & FAQ
435

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436 - **Two rallies with almost no gap?** Still two rows. `end_time[N] <= start_time[N+1]`,
never overlap.
437 - **Interrupted & replayed?** Label the interrupted attempt as its own row
(`ending_reason=let`), then
438 the replayed point as the next row.
439 - **Serve tossed but caught / re-served (no contact)?** No contact = no rally yet. Start
at the actual
440 serve contact; don't label the aborted toss.
441 - **Opponent reached the ball but netted/hit out?** That's a
`forced_error`/`unforced_error` (their
442 shot), not winner ? forced if your shot pressured them, unforced if it was routine.
443 - **Can't read the scoreboard?** Leave `score_before`/`score_after` blank. Never guess.
444 - **"Let" where the rules say play on?** If play continued, it's NOT a let ? keep one
rally and judge
445 the real ending.
446 - **Warm-up / knock-up / practice points?** Not real rallies ? do not label them. Start
at the first
447 real, scored serve. See the Rule 2 caveat (Part 2). When unsure whether the match has
started, flag
448 the clip ? don't guess.
449 - **Doubles?** Same rules ? a shot is any one racket/paddle contact by whichever player
hits it. Never
450 attribute `shots_count` to an individual, just total the contacts.
451 - **Server completely whiffs (no contact)?** Rare. True zero contact ? `shots_count` =
0`,
452 `ending_reason` = service_fault`, add a note. A graze = 1 contact = 1 shot.
453
454 ## Part 7 ? Self-check before you submit (per video)
455
456 - [ ] `rally_number` is 1,2,3? no gaps or repeats.
457 - [ ] Every `start_time` < `end_time` (check the auto-filled decimal columns populated).
458 - [ ] No overlaps: `end_time[N] <= start_time[N+1]`.
459 - [ ] Times entered as `m:ss.mmm`; decimal `start_time`/`end_time` auto-filled and look
right.
460 - [ ] No dead-time rows (scan the gaps ? dead time, not missing rallies).
461 - [ ] Every served point present, including service faults and lets.
462 - [ ] `shots_count` looks right (re-step any rally you rushed; doubles = team total).
463 - [ ] `ending_reason` in {winner, forced_error, unforced_error, service_fault, let,
other}; notes
464 filled when `other`.
465 - [ ] `last_shot_outcome` in {in, net, out, n_a} (`n_a` for service_fault/let).
466 - [ ] File downloaded as `.csv`, header row present, UTF-8.
467 - [ ] Did NOT trim / re-encode / rename the source video.
468 - [ ] **? (fusion)** intake-manifest row filled: `court_corners`, `net_line`,
`singles_or_doubles`.
469
470 ## Part 8 ? Mini glossary
471
472 - **Rally** ? one continuous passage of play from serve contact to the shuttle/ball
dead.
473 - **Shot** ? one contact between the shuttle/ball and a racket/paddle/bat (either
teammate in doubles).
474 - **Dead time** ? any time the shuttle/ball is not in active play. Never labeled.
475 - **Service fault** ? a failed/illegal serve that ends the point. Still a labeled rally.
476 - **Let** ? a replayed point. Labeled as its own rally with `ending_reason=let`.
477 - **Winner / forced_error / unforced_error** ? the final shot was a clean winner
(opponent made no
478 legal contact) / the loser's error was forced by a pressuring shot / the loser's error
was on a
479 routine, unpressured ball.
480 - **start_time / end_time** ? decimal seconds from the start of the provided file (auto-
filled from mm:ss).
481 - **? court_corners / net_line / singles_or_doubles** ? per-video calibration in the
intake manifest ($A).
482

```

```

483     ---
484
485     ## Appendix B ? Tier-B extensions (NOT in the day-1 deliverable) ? ADDED (fusion)
486
487     > Pilot these on the 3-video calibration round, then renegotiate price before scaling.
488     **Do not start
489     > these without explicit instruction** ? they are listed so Roora can quote them.
490     >
491     > - **Per-shot hit timestamps** ? a timestamp for *each* racket/paddle/bat contact (not
492     just the
493     > `shots_count` total). Extends the $7 CVAT method (box every contact, not only the
494     two boundary
495     > frames). Trains/validates the audio hit-detector and shot-cadence features.
496     > - **Per-shot type** ? for each contact, a label from {serve, clear, drop, smash, net,
497     drive, lift}.
498     > Needs its own short guideline + an inter-annotator agreement check.
499

```

15. assistant (2026-06-16T02:25:20.486Z)

Now the final doc ? QUALITY_ITERATIONS.md ? the empirical backbone. Let me read it in full.

16. user (2026-06-16T02:25:21.544Z)

```

1     # Rally-detection quality iterations ? living log
2
3     **Purpose.** A running, reverse-chronological log of how we evolve rally-detection
4     quality: each
5     iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson.
6     This is the
7     **live** continuation of the historical trail in
8     [`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the
9     archive policy
10    (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this
11    file stays at
12    `docs/` top level and keeps growing, then its entries graduate to the archive when they
13    conclude.
14
15    **Methodology (non-negotiable).** Every quality change rides the experiment harness
16    ([EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md)): new cues land **flag-gated, default
17    OFF**; they're
18    graded **offline against golden labels** (per-video + LOVO-honest); promotion to default
19    happens **only
20    on measured lift** through the `run_regression.py` gate; **rollback = flip a config
21    flag, never
22    `git revert`.** Cues are designed in
23    [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md); scoring
24    in [EVALUATION.md](EVALUATION.md); the 2-layer model in
25    [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md).
26
27    **Golden corpus = the 6 labelled matches** in `output/human_lovo_manifest.json` (owner-
28    confirmed
29    2026-06-13: use all 6 ? `adarsh_avi_singles, kushagra_singles, largetest_doubles,
30    gbaaddy,
31    testlarge_short, mahadevpura_singles`). The model trained on these is what we run
32    alongside the existing
33    detector on NEW footage for the side-by-side review.
34
35    ---
36
37    ## Baselines that drive strategy (IoU 0.5 vs human golden labels)
38
39    | Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
40    |---|---|---|
41    | Heuristic (velocity windowing) | 0.657 | 0.157 |

```



```

30 | Gen-0 owned head (LOVO, n=6) | 0.744 | 0.561 |
31 | Two-stage WASB+Gemini refine | 0.83 | ? |
32 | Gemini wrapper (`gemini-flash-latest`) | 0.93 | 0.66 |
33
34 **Reading:** clean footage is commoditized (serve via Gemini); **multi-court drops the
commodity wrapper
35 to 0.66 ? that is the owned model's ship target.** Baselines tracked in
`eval_baselines/`. The owner's
36 observed false positive ? a player **holding/flicking** the shuttle between points
counted as a rally ?
37 is the precision bug the fusion gates target.
38
39 ---
40
41 ## Current track (live) ? newest first
42
43 ### Served-side Gate-A measurement on the LIVE FusionSegmenter ? the independent run
behind the #151 revert *(2026-06-14)*
44 The owner-gated follow-up flagged by the (now **REVERTED**) SERVED-DEFAULT entry below.
An **independent** served-side check ?
45 separate from, and corroborating, the [#148](https://github.com/avidullu/badminton-
highlight-indexer/pull/148) litmus + #151's
46 run ? of whether the Gate-A default flip (PR #146) regresses the live path. **Verdict:
it does ? the flip was reverted in
47 [#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151); this entry is
the detailed measurement.** Driven the
48 **real** served seams (`_enrich_candidate_stats` ? `net_crossings`,
`_select_for_handoff` ? the gate) over each golden video's
49 **cached** WASB trajectory, at the **production windowing** `process_video` actually
uses
50 (`velocity_thresh=0.02, merge_gap=2.0, min_window=2.0`) ? no WASB re-run (the runner has
no cache-hit early-return), no GPU, no
51 Gemini. Config loaded exactly as production (`load_config()`); segmenters from the
registry. Repro:
52 `scratch/served_gateA_verify.py` + `scratch/served_gateA_sweep.py` (artifacts in
`output/gateA_served_verify/`); the standing
53 productionized guard is `backend/eval/served_gate_a.py` (#148).
54
55 - **(a) No-regression invariant CONFIRMED on real served code.** At `min_crossings = 0`,
`fusion_hybrid`'s seeding **and**
56 handoff set are **byte-identical to `wasb_hybrid`** on all 6 golden videos (same
candidate windows; handoff = all windows =
57 the base seam). The "fusion adds nothing when the gates are off" property is now
verified on the *live* served path, not
58 only by code review. ?
59 - **(b) The +0.074 win does NOT transfer to the served windowing.** The offline
promotion (+0.074, 6/6, p=0.031) was measured
60 on the eval harness's **recall-oriented *proposal* windowing** `(0.005, 1.0, 0.5)`,
where CONTROL **over-segments**
61 (segment-count ratio 1.68) and Gate-A's job is to clean up that fragmentation (?
1.01). The **served path uses a coarse
62 windowing** `(0.02, 2.0, 2.0)` that already merges aggressively (low over-
segmentation), so Gate-A has little FP to remove
63 and instead trims **real** rallies whose WASB trajectory registered only 0?1 net
crossings. **Candidate-handoff-level F1
64 (pre-Gemini, IoU ? 0.5) on the served windowing, n=6:**
65
66 | min_crossings | mean F1 | precision | recall | real-rally windows lost vs off | ?F1
vs off |
67 |---|---|---|---|---|---|
68 | 0 (Gate-A OFF = `wasb_hybrid`) | 0.300 | 0.317 | 0.290 | 0 | ? |
69 | 1 | **0.303** | 0.327 | 0.288 | 1 | **0.003** |
70 | **2 (shipped default, PR #146)** | 0.292 | 0.337 | 0.261 | **12** | **0.008** |
71 | 3 | 0.269 | 0.338 | 0.226 | 28 | ?0.031 |
72

```

73 At the shipped `mc=2`, mean F1 goes 0.300 ± 0.292 (± 0.008) and Gate-A drops **12**
 windows that match golden rallies**

74 (net_crossings almost all exactly 1.0), concentrated in doubles (`largetest\_doubles`
 ± 0.051). All deltas are **tiny** / within

75 $n=6$ noise**, so this is not a catastrophic regression ? but `mc=2` is **not** a
 served-side **improvement**, and the

76 held-shuttle "guard" is removing genuine rallies, not just feeds.

77 - **Served** relevance (honest). A dropped candidate window **never** reaches Gemini**, so
 each lost real rally is an

78 **unrecoverable** served recall loss; the precision "gain", by contrast, is partly
 what Gemini already does at the handoff ?

79 so the served-side tradeoff is at best neutral and plausibly net-negative, i.e. the
 candidate-level ± 0.008 likely

80 **understates** the served downside.

81 - **Root** cause = eval ? serve windowing. The harness promotes/refuses variants on a
 windowing the production path does not

82 serve. Gate-A's value is conditional on over-segmentation; the coarse served windowing
 doesn't over-segment, so the gate

83 inverts. [[decision-rigor-norm]]: the foundational issue (harness windowing ? served
 windowing) outweighs the single

84 proposal-windowing number.

85

86 **Outcome** ? REVERTED (see the entry below).

[#151](https://github.com/avidullu/badminton-highlight-indexer/pull/151)

87 restored the pre-#146 state: `config.json` `indexing.default\_segmenter` ? `gemini` (the
 `indexing.fusion` block removed) and

88 `FusionConfig.min\_crossings` default 2 ± 0 ? Gate A is now **OFF** by default but
 still opt-in**

89 (`indexing.fusion.min\_crossings: 2`), the persisted-superset invariant intact. Our
 independent numbers match #151's (mean

90 served ?F1 ± 0.008 , ~12 rallies dropped, 2/6 losses). If Gate A is ever opted back in on
 the served path, `mc=1` is the

91 least-bad operating point ($\pm$ flat F1, 1 rally lost). **Open** follow-up: **align** the
 experiment-harness windowing with the

92 served windowing** (or re-run promotions on the production windowing) so a harness lift
 actually predicts served behaviour ?

93 the real lesson here. The offline ablation + unit tests remain valid for what they
 measured.

94

95 **PERSONALIZATION** Phase-0 quality checkpoint ? the eval/safety RAILS for the per-user
 feature *(2026-06-14)*

96 The owner adopted the **HYBRID** posture: personalization as a **flag-gated** FEATURE on a
 generalization-first baseline**,

97 paired with a **contribution flywheel** so per-user gains and the shared baseline
 compound together (full design in

98 PERSONALIZATION_PLAN.md, merged

[#144](https://github.com/avidullu/badminton-highlight-indexer/pull/144)).

99 This checkpoint landed the three **honest-eval** RAILS** that keep a per-user/small-N
 tuning layer from promoting noise or

100 silently regressing a user:

101

102 - **Per-user** promotion gate ? [#141](https://github.com/avidullu/badminton-highlight-
 indexer/pull/141)**

103 (`backend/eval/per\_user\_gate.py`): a **min-N** floor** + **structural-guard** exemption**
 ? per-user/small-N tuning can't

104 promote noise and can't disable a structural guard (the proven Gate-A held-shuttle
 guard stays on regardless of a thin

105 user corpus).

106 - **Held-out-USER** learning curve ? [#142](https://github.com/avidullu/badminton-
 highlight-indexer/pull/142)**

107 (`backend/eval/per\_user\_eval.py`): turns the vague "<K videos to superior quality"
 claim into a **falsifiable** release

108 number** (the `smallest\_k\_with\_lift` measured on held-out users).

109 - **Per-user** no-regression guard ? [#143](https://github.com/avidullu/badminton-
 highlight-indexer/pull/143)**: a

```

110      retrain/baseline-upgrade can **never make a user worse than their incumbent** (frozen
per-user CONTROL; a regressing
111      candidate is refused).
112
113      **Read HONESTLY:** these are the **eval/safety rails** for the personalization feature ?
**not** new measured
114      rally-detection wins. No F1 moved this checkpoint; what landed is the discipline that
lets per-user tuning ship without
115      regressing the baseline or over-claiming on n=6-per-user data.
116
117      **Locked owner decisions (2026-06-14):** identity = *design for the north-star (multi-
user/cloud), implement for current
118      use (single-user local)*; the inline per-user classifier sits **behind a cloud-load
interface** (local store = one impl);
119      **`min_n` is a PROMOTION-confidence floor, NOT a service/segmentation gate** (a one-
video user is ALWAYS fully segmented by
120      the baseline batteries); **free-tier videos ARE pooled** into the shared baseline
(consent baked into free terms,
121      minors-exclusion enforced at the pooling query); and the previously-pending structural-
guard sub-decision is now LOCKED ?
122      **Gate-A-only structural + `min_n = 5`**.
123
124      ### Fusion P3 + P4 MEASURED on the n=6 golden corpus ? person = no lift, audio =
promising (not promotable) *(2026-06-14)*
125      The GPU person-feature extraction finished all 6 golden videos (`fusion_golden.py` ?
`output/*_golden_features.json`, the
126      replayable substrate); the CPU LOVO drivers then measured the lift with **honest stats
(C1 ? bootstrap 95% CI + paired sign
127      test; never a bare mean at n=6)**:
128
129      | Variant (honest LOVO, IoU ? 0.5) | F1 | ?F1 vs prior | 95% CI | sign test | CI clears
0 |
130      |---|---|---|---|---|
131      | shuttle-only | 0.307 | ? | ? | ? | ? |
132      | + person (P3) | 0.286 | **?0.021** (?6.7%) | [?0.165, +0.162] | 2W/4L (p=0.688) |
**No** |
133      | + person + audio (P4 incr.) | 0.366 | **+0.079** (+27.7%) | [?0.054, +0.203] | 4W/2L
(p=0.688) | **No** |
134
135      **Read honestly:** neither cue clears the promotion bar (both CIs span 0, both sign
tests n.s.) at n=6 ? **both stay
136      default-OFF** (promote-only-on-lift). **Person** adds noise (no court polygons ?
`players_in_court` counts spectators);
137      per-video it's a wash (testlarge +0.40, gbaaddy ?0.22). **Audio** is the more promising
lead ? +0.079 / +27.7%, wins 4/6
138      (largetest +0.27, mahadevpura +0.27), and shuttle+person+audio (0.366) lands **above**
shuttle-only (0.307) ? but n=6 can't
139      confirm it. **Next:** more matches (raise N until the sign test can resolve ~+0.08), and
measure audio on shuttle-only
140      **directly** (without the person handicap). Recorded in
`eval_baselines/experiment_leaderboard.json`. **Negative/uncertain
141      results are kept on purpose ? this is the honest ablation backbone for the paper aim.**
142
143      ### Gate-A served-default flip ? **REVERTED (eval?serve regression)** *(2026-06-14)*
144      **The #146 flip of Gate-A (net-crossings ? 2) to the live served default was REVERTED.**
A served-side check ?
145      the repeatable litmus in [#148](https://github.com/avidullu/badminton-highlight-
indexer/pull/148)
146      (`backend/eval/served_gate_a.py` + `tests/test_served_gate_a_regression.py`),
corroborated by an independent
147      served run ? found Gate A is **not a win on the served path**: on the production
(COARSE) windowing it is
148      **neutral-to-negative, mean served ?F1 ?0.008**, dropping ~12 real rallies across the
golden 6 (2 losses, 0
149      clear wins). It is a **confirmed served regression**, so the owner directed a revert to

```

```

restore the known-good
150     state.
151
152     **Why eval ? serve (the honest root cause).** The +0.074 below is a property of the
**offline harness's
153     permissive PROPOSAL windowing** (`velocity_threshold=0.005, merge_gap=1.0,
min_window=0.5`), which emits many
154     short candidate windows ? leaving abundant held-shuttle junk for Gate A to cut. The
**served COARSE windowing**
155     (`velocity_threshold=0.02, merge_gap=2.0, min_window=2.0`) already merges flight bursts
into a few long windows
156     and drops short ones (served over-seg ratio already **< 1**, i.e. under-segmentation),
so there is little junk
157     left to cut and the gate instead occasionally trims a borderline true window ? slightly
negative ?F1. The gate
158     *itself* is correct (#148's control test reproduces +0.074 / 6-of-6 / p=0.031 exactly
under proposal
159     windowing); the **operating point is what differs**. This is the canonical eval?serve
trap.
160
161     **What the revert restores** (back to the pre-#146 no-regression state):
162     - `config.json` ? `indexing.default_segmenter` **`fusion_hybrid` ? `gemini`** (the prior
served detector); the
163     explicit `indexing.fusion` block added by #146 is removed.
164     - `backend/config/models.py` ? `FusionConfig.min_crossings` default **2 ? 0** (Gate A
**OFF** by default).
165     - Tests (`tests/test_config.py`, `tests/test_fusion_hybrid.py`) restored to the Gate-A-
OFF defaults; the
166     **persisted-superset invariant is preserved** (gating only ever filtered the AI
handoff, never persistence).
167     - `eval_baselines/experiment_leaderboard.json` ? the #146 `promotion` record carries a
`reverted (eval?serve)`
168     note.
169
170     **Gate A is NOT deleted ? it remains opt-in** (`indexing.fusion.min_crossings: 2`) for
footage with more
171     held-shuttle/service-feed false positives than the golden 6, and the offline ablation
evidence below stands
172     (it is a real proposal-windowing win). The path back to serving it is to **align the
served windowing toward
173     the proposal operating point** and re-run the #148 litmus to confirm the lift transfers.
The #148 litmus is
174     now the standing guard: it pins the honest measured behaviour (served ~neutral; +0.074
only under proposal
175     windowing) and trips if a future windowing change silently revives or breaks either
fact.
176
177     ### Acting on the first A/B: Gate-A PROMOTED + learned variant FIXED + GX010125
corroboration *(2026-06-14)*
178     **Gate-A (net-crossings ? 2) ? PROMOTED (first cue promoted via the harness).** The
measured win
179     (?F1 +0.074, **6/6 golden matches, sign-test p=0.031**, CI [+0.042,+0.104]; over-seg
1.68?1.01 ? the
180     [first-A/B archive](archives/quality-iterations/2026-06-first-ab-experiment/README.md))
clears the honest
181     bar. **Recommended config:** `indexing.fusion.min_crossings = 2` on the fusion path (the
eval-only Gate A
182     folded into production, MULTI_SIGNAL_FUSION_PLAN §3.1). ? The offline win is on the
recall-oriented *proposal*
183     candidate set; flipping the **live served default** stayed owner-gated at the time of
this entry. **Update: the
184     flip was attempted (#146) and then REVERTED** ? on the served COARSE windowing Gate A is
neutral/negative
185     (?0.008), an eval?serve regression; see the REVERTED entry above. Gate A stays opt-in
186     (`indexing.fusion.min_crossings: 2`), default OFF. **Corroborated on real footage:**

```

```

offline
187     re-score of GX010125_1080p's *cached* trajectory (no GPU, isolated
`output/GX010125_run/`) ? 45 candidates ?
188     CONTROL 45 rallies vs **Gate-A 27**; Gate-A dropped **18 low-crossing windows (~1.7
min)** of likely
189     held-shuttle/feed FPS (no golden labels for this footage ? divergence, not F1).
190
191     **Learned variant FIXED ? [#130](https://github.com/avidullu/badminton-highlight-
indexer/pull/130).** The
192     first A/B's learned logistic lost to CONTROL and zeroed `testlarge` ? a fixed-0.5-cutoff
artifact. Added
193     additive/default-OFF `Variant.tune_threshold` (operating point chosen on the **train**
fold) + `calibrate`
194     (Platt/isotonic, C2). Validated on the n=6 golden set: **mean F1 0.307 ? 0.423
(+0.116)** , testlarge
195     0.000 ? 0.300, and the learned variant now **beats CONTROL (0.388)** . Threshold-in-LOVO
is the big lever;
196     calibration is available too.
197
198     **Next:** a real fusion-path run to confirm Gate-A served-side; per-video output layout
199     ([PER_VIDEO_OUTPUT_LAYOUT.md](PER_VIDEO_OUTPUT_LAYOUT.md)) so each video's artifacts are
self-contained.
200
201     ### First end-to-end A/B on the golden corpus (n=6) ? **ARCHIVED** *(2026-06-14)*
202     First real use of the harness + C1?C4 helpers, 100% offline (re-scored persisted WASB
trajectories, no GPU).
203     **Gate-A (net-crossings ? 2) is a measured win** ? ?F1 +0.074, 6/6 matches, sign-test
**p = 0.031** , CI
204     [+0.042, +0.104], over-segmentation (segment-count ratio) **1.68 ? 1.01** . The window-
level 5-feature logistic
205     **did not clear the bar** (?0.081, 2W/4L; a threshold/calibration artifact ? *not* the
stronger per-frame
206     prototype). The framework held up: **C1 promoted one variant and refused another; C4
surfaced over-segmentation
207     tIoU hid.** Full record + tables ?
208     [`archives/quality-iterations/2026-06-first-ab-experiment/`](archives/quality-
iterations/2026-06-first-ab-experiment/README.md).
209     Next: Gate-A is a candidate for the no-regression promotion gate; the logistic failure
is the C2 (calibration)
210     + threshold-in-loop task list.
211
212     ### Eval-honesty helpers (course corrections C1?C4) ? additive, default-OFF
*(2026-06-14)*
213     **Why.** A multi-agent literature review (folded into
214     [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) §13) confirmed the
framework's *skeleton* is
215     standard prior art (Temporal Action Localization/Segmentation + late/score-level fusion
? cite, don't claim),
216     and flagged that our **measurement** is walking into known small-N traps that would make
"promote only on
217     measured lift" promote noise or hide a real win. Four corrections, shipped as **pure,
additive helpers** (no
218     existing `compare`/LOVO/`metrics.score` behaviour changed ? callers opt in):
219     - **C1 ? never a single mean.** `backend/eval/eval_stats.py`:
`mean_with_ci`/`bootstrap_ci` (deterministic,
220     seeded), `paired_sign_test` (the exact distribution-free "won on enough *videos*?"
test the corpus already
221     needs), `paired_delta_ci` (B?A with a CI that must clear 0). At n?6 a lone LOVO mean
is not trustworthy.
222     - **C2 ? calibrate cue confidence.** `backend/eval/score_calibration.py`: Platt +
isotonic (PAVA) calibrators
223     + `brier_score`/`expected_calibration_error`, so a CNN softmax, an audio onset, and an
integer net-crossing
224     count are comparable before the arbiter and the over-confident cue can't swamp the
rest.

```

```

225     - **C3 ? honest stacking.** `eval_stats.out_of_fold_predictions`/`grouped_folds` (leave-
one-group-out by
226     match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no
leakage).
227     - **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`:
`f1_at_overlaps`
228     (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU),
`segment_count_ratio`. tIoU alone
229     is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler
exists to fix.
230
231     **Status.** Helpers + 24 unit tests landed (pure stdlib, run in the numpy/GPU-free
lanes); ruff + mypy clean.
232     **Not yet wired into the default harness flow** (wiring `compare`/`run_regression.py` to
report CIs + F1@k is
233     the follow-on, owner-gated). The discipline they encode is the methodology for every
future cue measurement ?
234     see §13/C1?C4 for the citations and the remaining (Tier-2/3) corrections.
235
236     ### Multi-signal fusion P2: the FusionSegmenter *(in progress, 2026-06-14)*
237     **Hypothesis.** Adjudicating shuttle-seeded windows with the net-crossing **Gate A**
(`min_crossings?2`)
238     and a person-occupancy **Gate B** raises **precision** on the held-shuttle / warm-up /
one-sided-feed /
239     spectator false positives ? *without* hurting recall, because the shuttle stays the
primary,
240     camera-invariant seed.
241
242     **Scope ? and an honesty boundary from the archive (read this).** The
243     [2026-06-multivideo-lovo](archives/quality-iterations/2026-06-multivideo-
lovo/README.md) iteration
244     already smoke-tested **court-region detection as a fix for the *multi-court contrast*
confound** and
245     found it **low-headroom**: only 12?14% of dead-time shuttle detections fall *outside*
the foreground
246     court, so a perfect court gate lifts contrast to only ~1.6?2.2× (below the 3.6× "works"
threshold) ? the
247     multi-court problem is **temporal**, and the **learned model**, not a court mask, is its
answer. So Gate B
248     here is **NOT** sold as the multi-court fix. It targets a **different, previously-
untested** failure mode
249     (held-shuttle/warm-up/one-sided/spectator FPS). Two distinct levers; don't conflate
them.
250
251     **Data support for Gate A.** The same archive shows real rallies on the multi-court
matches contain
252     **?2 net-crossings 97% of the time (median 5)** while a single service feed crosses
~once ? `min_crossings?2`
253     is a principled, data-backed discriminator (it's the held-shuttle owner-bug killer).
254
255     **Shipping shape.** A NEW `fusion` segmenter, **default OFF** (`default_segmenter`
unchanged); Gate A
256     wired into the live trajectory path behind `indexing.min_crossings` (default 0 = OFF ?
wasb/tracknet
257     byte-identical today); the court mask is an **optional** config polygon (omit ? Gate B
is
258     player-count-only; supply ? masked). Person features (`players_in_court,
two_sided_motion,
259     mean_player_motion, in_court_ratio, shuttle_player_colocation`) persisted into candidate
`stats` for P3's
260     learned fusion. Reuses `PersonDetector` (P1) + `rally_gate` (already existed) ? no
reinvention.
261
262     **Status (2026-06-14): SHIPPED, default-OFF.** `FusionSegmenter`
(`backend/pipeline/segmenters/fusion_hybrid.py`,

```

```

263     registry key `fusion_hybrid`) + the pure feature math
(`backend/pipeline/detectors/fusion_features.py`) + two
264     identity hooks on the trajectory engine (`_enrich_candidate_stats`,
`_select_for_handoff`) + `PersonDetector`.
265     detections_per_frame` + the `FusionConfig` block. Built design-first (a
`fusion-p2-design` fan-out workflow), then
266     **adversarially reviewed by a 31-agent workflow (25 findings ? 19 confirmed)**: the
headline confirmation is
267     **NO-REGRESSION verified** ? wasb/tracknet twins persist byte-identical stats and hand
off all windows (pinned by
268     the extended equivalence test); default `FusionSegmenter` == its wasb substrate. The
confirmed low/medium findings
269     were fixed (frame-dim guards, `round()` frame indices, the per-window axis-estimate
redundancy ? a reusable
270     `rally_gate` estimate, honest torch-import docstrings, and 6 new tests closing the
person-detector / Gate-B /
271     substrate / colocation-halo gaps). Suite 441?467 green**; ruff + mypy clean.
272     **Two findings deferred with intent** (reviewer-agreed): person features are computed
eagerly even for
273     Gate-A-pruned windows (a P3 redesign, would change the persisted-superset semantics);
and `fusion.velocity_threshold`
274     must be set equal to the substrate's to A/B against a tuned wasb run (documented in
`FusionConfig`).
275     **Lift still to be measured** offline on the 6 golden videos via
`experiment.compare(CONTROL, fusion)`; the
276     held-shuttle-drop + person-feature values are GPU/owner-side. *(Measured numbers
appended here when run ? no claims before.)*
277
278     ### Person cue extracted to a reusable producer (fusion P1) ?
[#115](https://github.com/avidullu/badminton-highlight-indexer/pull/115), 2026-06-13
279     Lifted the torchvision detector + ByteTrack + velocity windowing out of `yolo_hybrid`
into
280     `backend/pipeline/detectors/person_detector.py::PersonDetector`. **No behavior change**
(proven by the
281     unchanged `process_video` output tests); `yolo_hybrid` now orchestrates only. **Why it's
a quality move:**
282     it makes "add the person cue to fusion" an *add-a-producer*, not a segmenter rewrite.
Suite 440?441 green.
283
284     ### Experiment harness: A/B without rollback risk ?
[#114](https://github.com/avidullu/badminton-highlight-indexer/pull/114), 2026-06-13
285     Shipped `backend/eval/experiment.py` ? the thin variant/experiment layer (the audit's
"~80% already
286     exists" held; pure glue over
`metrics`/`calibration`/`classifier`/`training`/`regression.gt_hash`). Gives
287     `Variant` (+ pinned `CONTROL`), `compare()` (labeled A/B, LOVO-honest),
**`compare_unlabeled()`** (the SxS
288     read on NEW unlabeled footage ? agreed / A-only / B-only), and a JSON experiment
leaderboard. **This is
289     the instrument every iteration is measured on.** 19 tests incl. the no-regression
invariant (all cues OFF
290     ? byte-identical to control). Suite 421?440 green.
291
292     ---
293
294     ## Archived predecessors (the historical trail)
295     Completed iterations live under [archives/quality-iterations/`](archives/quality-
iterations/README.md):
296     - **`2026-06-initial/`** ? seed of the quality phase.
297     - **`2026-06-human-gt-tuning/`** ? first eval vs **human** GT on a partial-label video;
partial-label
298     harness + labeled-window methodology; baseline 0.650 ? tuned 0.742 (*fit-on-self*);
found+fixed a
299     negative-start ingestion bug (collector PR #13).
300     - **`2026-06-multivideo-lovo/`** ? the **n=6 corpus + honest LOVO**; established the

```

```

heuristic is
301     **footage-fragile** (multi-court Kushagra 0.16 / gBaaddy 0.28), the ***"contrast"
metric** predicts
302     heuristic F1 monotonically, and a **learned per-frame prototype overtook the heuristic
at n=3 and holds
303     ~+0.10** (n=6: learned 0.583 vs heuristic 0.480), with **AUC 0.83?0.92 decoupled from
contrast**. This
304     is the data-backed green light for the owned model.
305
306     ---
307
308     ## Standing lessons (carry-forward)
309     - **Persist detection once, re-score variants forever.** Expensive/risky = per-frame
detection (GPU/WSL);
310     cheap/swappable = the decision. Most experiments never enter the served pipeline.
311     - **LOVO or it didn't happen.** Score on a held-out *video*, never a held-out window
from the same video.
312     Group by **match**, not file. Sign test needs n ? ~n=10/9-wins for p<0.05.
313     - **The multi-court confound is TEMPORAL, not spatial/loudness** ? three heuristics were
tested and failed
314     for the same reason (foreground spatial gate **?0.04**; audio energy **AUC ~0.6**;
court-region
315     separability **12?14% out-of-region**): the dead-time confusion is rally-vs-non-rally
activity in the
316     *same place and frequency band*. Only the learned model's temporal/structural features
crack it (AUC
317     0.83 on Kushagra/gBaaddy). **Don't re-derive these dead ends.**
318     - **Court-masking ? multi-court fix** (see above) ? it's a **precision** tool for off-
court persons and
319     one-sided feeds, a *different* lever; measure its lift, don't assume it.
320     - **Net-crossings are a strong rally signal** ? 97% of real rallies cross ?2x (median
5); single feeds ~1x.
321     Gate A leans on this.
322     - **Small-N feature discipline.** Few, scale/translation-invariant features
(counts/ratios, not raw
323     coords) + a logistic regression, not a net ? raw coords memorize this court/camera.
324     - **Calibration, not representation, is the small-N wall** ? learned probabilities
separate (high AUC) but
325     the operating point doesn't transfer with few training videos. More distinct matches
is the highest-ROI
326     unblock.
327     - **Gate A already existed, just wasn't wired to production** ? the cheapest precision
win is plumbing, not
328     new capability. Always check what the eval drivers already do that the live path
doesn't.
329

```

17. assistant (2026-06-16T02:25:24.640Z)

I have all nine docs read in full. Let me check a couple of referenced artifacts to ground the answer about what is actually committed to the repo (the crux question) versus gitignored.

18. user (2026-06-16T02:25:27.720Z)

```

=== eval_baselines/ (tracked?) ===
eval_baselines/experiment_leaderboard.json
eval_baselines/gemini_wrapper_2026-06-10.json
eval_baselines/heuristic_lovo_n6.json
=== eval_baselines dir listing ===
total 28
drwxr-xr-x 1 avidu 197609    0 Jun 14 16:28 .
drwxr-xr-x 1 avidu 197609    0 Jun 16 01:52 ..
-rw-r--r-- 1 avidu 197609 4696 Jun 14 16:28 experiment_leaderboard.json
-rw-r--r-- 1 avidu 197609 1977 Jun 10 17:43 gemini_wrapper_2026-06-10.json
-rw-r--r-- 1 avidu 197609  689 Jun 10 15:57 heuristic_lovo_n6.json

```



```
=== gitignore golden/output refs ===
output/foo_golden_features.json
output/x.db
=== .gitignore relevant lines ===
Binary file .gitignore matches
```

19. assistant (2026-06-16T02:25:31.084Z)

The `output/` dir is gitignored (golden features + DB live there, per-machine), but `eval\_baselines/` IS tracked. Let me inspect the tracked baseline/leaderboard files ? these are the structured per-video artifacts already committed to the repo, which is central to the crux question.

20. user (2026-06-16T02:25:31.593Z)

```
1      [
2      {
3          "timestamp": "2026-06-14T06:23:51.507393+00:00",
4          "iou": 0.5,
5          "label_set_hash": "50d98ffbc370",
6          "num_videos": 6,
7          "variant_a": {
8              "name": "shuttle_only",
9              "spec_hash": "47c867c36f2f",
10             "aggregate": {
11                 "precision": 0.31807603780927995,
12                 "recall": 0.3101947096042718,
13                 "f1": 0.3068611057861412,
14                 "mean_iou": 0.5791831889148886
15             }
16         },
17         "variant_b": {
18             "name": "shuttle_person_fusion",
19             "spec_hash": "253f4709b4c4",
20             "aggregate": {
21                 "precision": 0.3617228001510626,
22                 "recall": 0.2708939086387243,
23                 "f1": 0.2863398846047612,
24                 "mean_iou": 0.5690111885714347
25             }
26         },
27         "delta": {
28             "precision": 0.04364676234178266,
29             "recall": -0.0393008009655475,
30             "f1": -0.02052122118138,
31             "mean_iou": -0.01017200034345389
32         },
33         "honest_delta_f1": {
34             "mean_delta": -0.02052122118137999,
35             "ci_low": -0.16462806145542527,
36             "ci_high": 0.16224446866060574,
37             "n": 6.0,
38             "ci_excludes_zero": false,
39             "sign_test": {
40                 "wins": 2.0,
41                 "losses": 4.0,
42                 "ties": 0.0,
43                 "n_effective": 6.0,
44                 "p_value": 0.6875
45             }
46         }
47     },
48     {
49         "timestamp": "2026-06-14T06:24:56.487711+00:00",
50         "iou": 0.5,
```

```

51     "label_set_hash": "50d98ffbc370",
52     "num_videos": 6,
53     "variant_a": {
54         "name": "shuttle_person",
55         "spec_hash": "ee89f1e41604",
56         "aggregate": {
57             "precision": 0.3617228001510626,
58             "recall": 0.2708939086387243,
59             "f1": 0.2863398846047612,
60             "mean_iou": 0.5690111885714347
61         }
62     },
63     "variant_b": {
64         "name": "shuttle_person_audio",
65         "spec_hash": "d68c39eba87c",
66         "aggregate": {
67             "precision": 0.4530798177857001,
68             "recall": 0.3408138120839734,
69             "f1": 0.3655826552797436,
70             "mean_iou": 0.7074802852140344
71         }
72     },
73     "delta": {
74         "precision": 0.0913570176346375,
75         "recall": 0.06991990344524907,
76         "f1": 0.07924277067498242,
77         "mean_iou": 0.13846909664259965
78     },
79     "honest_delta_f1": {
80         "mean_delta": 0.07924277067498244,
81         "ci_low": -0.053508315003560535,
82         "ci_high": 0.2029045239922802,
83         "n": 6.0,
84         "ci_excludes_zero": false,
85         "sign_test": {
86             "wins": 4.0,
87             "losses": 2.0,
88             "ties": 0.0,
89             "n_effective": 6.0,
90             "p_value": 0.6875
91         }
92     }
93 },
94 {
95     "timestamp": "2026-06-14T00:00:00+00:00",
96     "iou": 0.5,
97     "label_set_hash": "50d98ffbc370",
98     "num_videos": 6,
99     "promotion": {
100         "cue": "gate_a_net_crossings",
101         "promoted_to": "served_default",
102         "decision": "PROMOTED ? Gate A (net_crossings>=2) flipped to the live served
default",
103         "reverted": "reverted (eval?serve): the #146 served flip was REVERTED ? on the
served COARSE windowing Gate A is neutral/negative (mean served ?F1 -0.008, ~12 real rallies
dropped), confirmed by the #148 served litmus + an independent served check. The +0.074 below is
real only on the harness's recall-oriented PROPOSAL windowing; it does NOT transfer to the
served operating point. Restored config.json default_segmenter=gemini and
FusionConfig.min_crossings default=0. Gate A stays opt-in (indexing.fusion.min_crossings=2).",
104         "served_change": "config.json indexing.default_segmenter gemini->fusion_hybrid;
FusionConfig.min_crossings default 0->2",
105         "reversible": "indexing.fusion.min_crossings=0 (Gate-A rollback) or
indexing.default_segmenter=wasb_hybrid (full revert)",
106         "gate_b": "stays OFF (min_players=0, person cue OFF)",
107         "owner_approved": true,

```

```

108         "real_video_corroboration": "GX010125_1080p offline re-score: 45 candidates ->
CONTROL 45 rallies vs Gate-A 27 (dropped 18 low-crossing held-shuttle/feed windows, ~1.7min)",
109         "over_seg_segment_count_ratio": {"before": 1.68, "after": 1.01},
110         "source_archive": "docs/archives/quality-iterations/2026-06-first-ab-
experiment/README.md"
111     },
112     "variant_a": {
113         "name": "control_shuttle_proposals",
114         "spec_hash": "control",
115         "aggregate": {}
116     },
117     "variant_b": {
118         "name": "gate_a_min_crossings_2",
119         "spec_hash": "gateA_mc2",
120         "aggregate": {}
121     },
122     "delta": {
123         "f1": 0.074
124     },
125     "honest_delta_f1": {
126         "mean_delta": 0.074,
127         "ci_low": 0.042,
128         "ci_high": 0.105,
129         "n": 6.0,
130         "ci_excludes_zero": true,
131         "sign_test": {
132             "wins": 6.0,
133             "losses": 0.0,
134             "ties": 0.0,
135             "n_effective": 6.0,
136             "p_value": 0.031
137         }
138     }
139 }
140 ]

```

21. user (2026-06-16T02:25:32.053Z)

```

1      {
2          "name": "heuristic_velocity_windowing_LOVO",
3          "description": "The honest cross-video floor the Gen-0 learned model must beat:
calibrated velocity-windowing heuristic, leave-one-video-out over the n=6 owned golden corpus.",
4          "computed": "2026-06-10, reproduced via: python -m backend.eval.calibrate_local
--manifest output/human_lovo_manifest.json --metric f1 (matches docs/archives/quality-
iterations/2026-06-multivideo-lovo)",
5          "mean_f1": 0.480,
6          "std": 0.242,
7          "per_video": {
8              "adarsh_avi_singles": 0.751,
9              "kushagra_singles": 0.157,
10             "largetest_doubles": 0.737,
11             "gbaaddy": 0.309,
12             "testlarge_short": 0.267,
13             "mahadevpura_singles": 0.657
14         }
15     }
16

```

22. assistant (2026-06-16T02:25:35.628Z)

This is the crux finding. Let me check the regression baseline path referenced in the harness doc (`eval\_baselines/regression\_baseline.json`) and whether the golden label CSVs themselves are tracked anywhere, plus the manifest.

23. user (2026-06-16T02:25:39.660Z)

```
=== regression_baseline.json tracked? ===
docs/archives/checkpoints/2026-06-eval-baseline/README.md
eval_baselines/experiment_leaderboard.json
eval_baselines/gemini_wrapper_2026-06-10.json
eval_baselines/heuristic_lovo_n6.json
=== any golden CSV / rallies.csv tracked? ===
backend/eval/fusion_golden.py
docs/GOLDEN_DATA_SHARING.md
docs/GOLDEN_SET_IMPLEMENTATION.md
docs/GOLDEN_SET_PHASE1_VIDEOS.md
docs/GOLDEN_SET_VENDORING.md
tests/test_fusion_golden_telemetry.py
=== golden_features mention in tracked code ===
backend/eval/fusion_golden.py
backend/eval/served_gate_a.py
docs/PER_VIDEO_OUTPUT_LAYOUT.md
tests/test_fusion_golden_telemetry.py
tests/test_served_gate_a_regression.py
=== PER_VIDEO_OUTPUT_LAYOUT doc exists? ===
-rw-r--r-- 1 avidu 197609 20087 Jun 15 14:11 docs/MULTI_SIGNAL_FUSION_PLAN.md
-rw-r--r-- 1 avidu 197609 4408 Jun 14 11:42 docs/PER_VIDEO_OUTPUT_LAYOUT.md
```

24. assistant (2026-06-16T02:25:53.099Z)

No golden label CSVs, no `*_golden_features.json`, and no `regression_baseline.json` are tracked in git. The `output/` substrate is gitignored and shared via OneDrive only. The `eval_baselines/*.json` (scores/leaderboard, no raw labels) ARE tracked. Let me read the `fusion_golden.py` header and `PER_VIDEO_OUTPUT_LAYOUT.md` to pin the exact `*_golden_features.json` shape and confirm no doc proposes committing per-video label/feature artifacts to the repo.

25. user (2026-06-16T02:25:54.153Z)

```
1      # Per-video output layout ? one directory per video (design / plan)
2
3      **Status:** **DESIGN ? owner-gated, NOT started.** Proposed convention so processing N
videos
4      yields N self-contained subdirectories instead of one tangled `output/`. Docs-only.
5
6      ## The problem
7      Today every run writes into a **flat `output/**` with filename prefixes ? highlights
8      (`<video>_highlights.mp4`), run logs (`<video>_run.log`), eval CSVs
9      (`<video>_vs_human.csv`),
10     the shared SQLite DB (`sports_indexer.db`), plus **hardcoded shared subdirs** the CLI
11     `--output-dir` does **not** redirect:
12     - `output/wasb/` and `output/tracknet/` ? trajectory CSVs + frame caches
13     - `trajectory_hybrid.py:~176`, `out_dir = os.path.join("output", self.family)` ?
literal `output`.
14     - `output/chunks_{provider}_handoff/` ? the AI-handoff clip staging (same file).
15
16     So 5 videos produce an interleaved pile, and `--output-dir` only relocates *some*
artifacts
17     (highlights/clips/DB) while trajectories and handoff clips still land in the shared
`output/`.
18     The first isolated run (`output/GX010125_run/`, 2026-06-14) had to be done by hand and
is only
19     *partially* isolated for this reason.
20
21     ## The target
22     ```
23     output/
24         <video_slug>/                                # slug = sanitized stem + short video_id (collision-
safe)
```

```

24         runlog.txt                # the per-video run log (today: output/<video>_run.log)
25         sports_indexer.db          # per-video DB (or a row-scoped view of a shared DB ?
see below)
26         wasb/<key>_wasb.csv         # the trajectory (today: output/wasb/)
27         handoff/                   # AI-handoff clips (today: output/chunks*_handoff/,
auto-cleaned)
28         rallies.csv                # detected rally [start,end] + shot_count
29         segments/                  # per-segment clips / pngs / debug frames
30         highlights.mp4              # the reel
31         report.json / report.md     # the observability report
32         ...
33     A directory with 5 videos ? 5 subdirectories, each holding everything for that
video.
34
35     ## Design
36     - video_slug(video_path, video_id) -> str ? a new helper (next to
`utils/hashing.py`):
37         sanitized stem + `video_id[:12]` (the MD5 already used as the cache key), e.g.
38         `GX010125_1080p__89605e80ed01`. Collision-safe (same stem / different bytes ?
different dir).
39     - Thread one `run_dir` through the pipeline instead of the literal "output":
40     - TrajectoryHybridSegmenter.process_video ? out_dir = os.path.join(run_dir,
self.family)
41         and the handoff dir under run_dir (replaces the two hardcoded "output" joins).
42     - The wasb_runner / tracknet_runner already take output_win_dir ? just pass the
per-video one.
43     - The CLI --output-dir (already in backend/main.py) becomes the parent; the
pipeline
44         creates <output-dir>/<video_slug>/ and writes everything under it.
45     - DB choice (owner decision): (a) a per-video sports_indexer.db in the slug dir
(simplest,
46         fully isolated, but loses cross-video queries), or (b) keep the shared DB and only
isolate
47         files (the slug dir holds artifacts; the DB stays one queryable store keyed by
`video_id`).
48     - Recommend (b) ? the DB is already video_id-scoped; isolate the heavy file
artifacts,
49         keep one queryable metadata store. The slug dir then omits its own DB.
50     - Back-compat: default parent stays output/; a flag (or just the default-on
behaviour)
51         switches flat?per-video. Old flat outputs are left untouched; new runs go to slug
dirs.
52
53     ## Phasing (owner-gated; ONE PR per slice, off `master`)
54     - L0 ? this plan.
55     - L1 ? video_slug helper + thread run_dir through trajectory_hybrid (replace
the two
56         hardcoded "output" joins) + the per-video subdir creation; CLI --output-dir =
parent.
57         Pure plumbing; a regression test pinning that all artifacts land under the slug dir.
58     - L2 ? runlog + report + rallies.csv into the slug dir; migrate the eval/CSV
writers.
59     - L3 ? cache-hygiene + cleanup_caches awareness of the new layout (per-video purge).
60
61     ## Relation to the first isolated run
62     output/GX010125_run/ (the 2026-06-14 offline re-score) is the manual prototype of
this
63     layout ? it isolated the rally CSVs + summary, but the trajectory still lives in the
shared
64     output/wasb/ precisely because of the hardcoded path L1 fixes. This plan makes that
isolation
65     automatic and complete for every video.
66

```

```

1      """Offline golden-set person-feature extractor ? the fusion-P3 measurement's GPU step.
2
3      Mirrors `distill_local`'s CSV-driven flow but ADDS the ~5 person-cue features per
shuttle
4      candidate window, by running `PersonDetector` over the ORIGINAL video. It reuses the
5      EXISTING golden trajectory CSVs (so **no WASB re-run**), needs **no Gemini and no DB**,
and
6      writes one small JSON per video. The LOVO comparison (`fusion_compare.py`) then runs
100% on
7      CPU offline. So the GPU touches exactly one thing ? person detection ? and the lift is
8      measured by the shipped experiment harness; person features enter only as feature
columns the
9      learned scorer weights (no special-casing, default-OFF until the golden ?F1 justifies
them).
10
11      Run (on the GPU box):
12          python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json
--out-dir output
13
14      Each output `output/<name>_golden_features.json` is the replayable feature substrate ?
re-run
15      `fusion_compare` with different variants/thresholds forever without re-detecting on the
GPU.
16      """
17      from __future__ import annotations
18
19      import argparse
20      import json
21      import os
22      import time
23      from typing import Any, Dict, List, Optional
24
25      from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
26      from backend.pipeline.detectors import rally_gate
27      from backend.pipeline.detectors import fusion_features as ff
28      from backend.eval import distill_local
29      from backend.eval.calibrate_wasb import load_golden_gts
30      from backend.eval.training import label_candidates
31      from backend.utils.run_telemetry import RunTelemetry
32
33      # The 5 fps/resolution-invariant shuttle features distill_local already computes.
34      SHUTTLE_FEATURE_NAMES = list(distill_local.FEATURE_NAMES)
35      # The 5 person-cue features (fusion_features.PERSON_FEATURE_NAMES).
36      PERSON_FEATURE_NAMES = list(ff.PERSON_FEATURE_NAMES)
37
38
39      def _derive_video_path(spec: Dict[str, Any]) -> str:
40          """Original MP4 for a manifest entry ? explicit `video_path`, else derived from the
41          golden labels path (`X.rallies.csv` -> `X.MP4`, the golden-set naming
convention)."""
42          if spec.get("video_path"):
43              return str(spec["video_path"])
44          labels = str(spec["labels"])
45          if labels.endswith(".rallies.csv"):
46              return labels[: -len(".rallies.csv")] + ".MP4"
47          raise ValueError(f"{spec.get('name')}: no video_path and labels {labels!r} isn't
*.rallies.csv")
48
49
50      def extract_video(spec: Dict[str, Any], detector: Optional[Any] = None,
51                      frame_skip: int = 3, iou: float = 0.5,
52                      log_every_sec: float = 15.0,
53                      telemetry: Optional[Any] = None) -> Dict[str, Any]:
54          """One golden video -> a feature record (shuttle + person features per candidate
window).

```

```

55
56     Returns ``{name, fps, frame_width,
candidates:[{start,end,shuttle{,person{,label}}, gts}``.
57     `detector` (anything with `detections_over_windows`) is injectable for GPU-free
tests;
58     when None a real `PersonDetector` is built (loads torchvision on first use).
59     `telemetry` (a `RunTelemetry` or None) records per-frame velocity + machine-health
snapshots
60     from the detection progress callback so a hard kill leaves a forensic runlog.
61     ""
62     fps, fw = float(spec["fps"]), float(spec["frame_width"])
63     traj = str(spec["trajectory"])
64     points = TrackNetRunner.parse_trajectory_csv(traj)
65     intervals, x_shuttle = distill_local.build_candidates(traj, fps, fw)
66     gts = load_golden_gts(str(spec["labels"]))
67     labels = label_candidates(intervals, gts, iou)
68     # Net axis for the person two-sided split ? reuse rally_gate's camera-agnostic
estimate
69     # (parity with the shuttle net-crossing gate) rather than hard-coding an
orientation.
70     axis, net_px, _span = rally_gate.estimate_play_axis(points)
71     video_path = _derive_video_path(spec)
72
73     if detector is None:
74         from backend.pipeline.detectors.person_detector import PersonDetector
75         detector = PersonDetector({"indexing": {"use_gpu": True}})
76
77     # Single-pass person detection over ALL candidate windows (open the video ONCE, read
78     # forward ? no per-window seek). Then compute features per window (pure, no GPU).
79     n = len(intervals)
80     win_frames = [(round(s * fps), round(e * fps)) for (s, e) in intervals]

```

27. assistant (2026-06-16T02:26:00.724Z)

I now have everything needed to answer precisely, including the exact `\*\_golden\_features.json` shape. Let me compose the structured output.

The crux finding is clear: **no doc** proposes committing structured per-video label/feature artifacts to the repo for video-free regression. The closest existing mechanism is `GOLDEN\_DATA\_SHARING.md`, which deliberately keeps those artifacts OUT of git (OneDrive sync of gitignored `output/`). What IS committed is the small score/leaderboard JSONs in `eval\_baselines/` (aggregate + per-video F1, no raw labels/features). This is the gap a new proposal would extend.

28. user (2026-06-16T02:27:54.920Z)

Structured output provided successfully

29. assistant (2026-06-16T02:28:01.191Z)

I've completed the analysis of all nine docs (plus the two cross-referenced ones, PER_VIDEO_OUTPUT_LAYOUT.md and the tracked eval_baselines/ artifacts, to ground the crux question) and returned the findings via StructuredOutput.

The headline for the crux question: **no doc** proposes committing structured per-video label/feature artifacts to the repo for video-free regression. The only existing per-video-artifact sharing design (GOLDEN_DATA_SHARING.md) deliberately does the opposite ? keeps them gitignored in `output/` and syncs via a private OneDrive folder because they derive from the proprietary corpus. What the repo already commits is only the small score/leaderboard JSONs in `eval\_baselines/` (aggregate + per-video F1, deltas, honest CI/sign-test, and a `label\_set\_hash`/`gt\_fingerprint` that binds a baseline to a label set without storing the labels). So a new proposal would extend an unfilled gap, not duplicate or contradict an existing one ? provided it reconciles with the licensing/privacy guardrails (proprietary-derived data, minors and biometric exclusions) and the OneDrive convention.